Supplementary Material for "RWMMO: A Real-World Benchmark for Multimodal Multi-Objective Optimization with Computable Modality Certificates"

This supplement provides per-problem details (S1), a per-problem equivalent-design showcase with two optima side by side (S2), the reference-archive layout and variable semantics (S3), the licence and provenance audit (S4), independent oracles (S5), the full statistical tables of the eval_v2 study (S6), trajectory grids and gallery indexes (S7), MMI robustness studies (S8), and reproduction instructions (S9). All tables labelled "auto-generated" are produced by scripts from the on-disk artefacts and re-validated by the project's regression suite; no number in this supplement is typed by hand.

S1. Per-Problem Details

Table S1 lists, for each of the 20 official problems, the certificate numbers that the main paper summarises only in aggregate: decision/objective dimensions, encoding, reference tier, modality label, MMI and $\mathrm{MMI}_{\mathrm{far}}$, exact tie-group count, maximum multiplicity, archive row counts (global front and published local certificate), and the licence status. The profile blocks below the table add, per problem, the key numbers in context, the machine-maintained English evidence sentence from the official manifest, and the archive file names that ship with the release. The authoritative per-problem narrative (constraint semantics, objective definitions, oracle identities, evidence paths) is maintained in the benchmark's machine-generated problem catalogue, which ships with the release and is byte-compared against the metadata by the regression suite; both Table S1 and the profiles are generated from the same three JSON sources (official manifest, candidate registry, modality certificates) plus the visual-summary and initial-distribution CSVs.

Three reading notes. First, the four GLPF problems (RWMMOP11, RWMMOP12, RWMMOP13, RWMMOP14) have $\mathrm{MMI} = 0$ *by complete enumeration*: their fronts are realised by one connected decision region each; their trap-axis evidence lives in the local-certificate column. Second, the three †-marked problems of the main paper's Table I (RWMMOP1, RWMMOP2, RWMMOP9) cap tie multiplicity at 8 in the archive, so both their MaxMult and their MMI are lower bounds. Third, RWMMOP13 is the deliberate anchor problem (smallest instance, retained to calibrate basic behaviour); its difficulty tier is recorded as anchor rather than hard.

RWMMOP1 — SatelliteTTC25 (satellite scheduling)

D / M / encoding:	164/2/integer
Reference tier:	high-budget-approx
Modality label:	EM
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	4.27 / 4.27†
Tie groups / max multiplicity:	35 / 8†
Front / local rows:	181 /
Max decision separation:	5.729

Evidence: Satellite TT&C scheduling on the frozen case_25 instance (25 requests over the request x pass x equipment tensor, compacted to D=164 integer slots, M=2: negative executed priority reward and finish time). The wrapper reproduces the vendored evaluator field by field in a three-source differential (wrapper versus project copy versus vendored original, maximum absolute difference 0.0). The archive was rebuilt on 2026-09-17 by a three-stage union sampler - 16000 uniform and class-initialiser schedules plus 512 constructed equivalent partners, all of them accepted at bit-identical objectives (largest rejected deviation 0) - giving 181 rows over 35 distinct objective vectors, 35 exact tie groups, largest group 8. The equivalence is structural and measured, not coincidental: requirements 19, 20 and 21 are executed by no schedule in 16000 samples, so their equipment slots cannot reach any objective, and re-randomising the equipment of requirements that are merely unexecuted in one schedule is NOT neutral (it changes who wins the equipment conflict; the first attempt moved the makespan by 111128 s), which is why every partner is verified by the evaluator before it is accepted. Per-objective multiplicity is capped at 8 so IGD and IGDX are weighted by front geometry rather than by duplication (the uncapped maximum was 465, attracted by the degenerate do-nothing schedule whose reward-0 / finish-0 point is genuinely non-dominated); the uncapped value stays in the archive metadata. Every archived row is re-evaluated from its own decision vector before writing (deviation 0). The legacy evaluator returns no constraint vector, so feasibility appears as unexecuted requests and every formal result must report executedRequests next to the objectives (mean 10.2351875 of 25 executed, no sampled schedule executed all 25). Objectives are normalised with constants frozen into SatelliteTTC25_full.mat

Table S1: Per-problem certificate numbers (auto-generated by `write_paper_tables.m`). TieG: exact tie groups on the published front; MaxMult: maximum tie multiplicity; Front/Local: published global / local archive rows; Lic.: licence status.

ID	Class	D	M	Enc.	Tier	Label	MMI	$\mathrm{MMI}_{\mathrm{far}}$	TieG	MaxMult	Front	Local	Lic.
RWMMOP1	SatelliteTTC25	164	2	integer	approx	EM	4.27	4.27	35	8	181		verified
RWMMOP2	Jobshop5	36	2	permutation/integer	approx	EM	5.84	5.84	163	8	1041		verified
RWMMOP3	SatelliteTTC50	352	2	integer	approx	EM	0.62	0.62	3	4	48		verified
RWMMOP4	ArrayNull	8	2	real	approx	EM	1.00	1.00	74	2	148		verified
RWMMOP5	WindFarm	16	2	real	approx	EM	7.00	7.00	13	8	104	104	verified
RWMMOP6	NestingLayout	18	2	real	approx	EM	6.96	6.96	57	8	452	588	verified
RWMMOP7	EcoDispatch15MF	15	2	real	approx	EM	1.35	1.35	74	8	269		verified
RWMMOP8	Knapsack24	24	2	binary	exact	EM	7.76	7.76	147	27	924		verified
RWMMOP9	WTA66	36	2	integer	approx	EM	6.95	6.95	33	8	258		verified
RWMMOP10	Truss25	25	2	real	approx	EM	0.92	0.92	136	2	295		verified
RWMMOP11	DNR69	73	3	binary	exact	GLPF	0.00	0.00	0	1	6	1529	verified
RWMMOP12	DNR33MS	37	3	binary	exact	GLPF	0.00	0.00	0	1	12	337	verified
RWMMOP13	DNR33MP	2	3	integer	exact	GLPF	0.00	0.00	0	1	7	50	verified
RWMMOP14	Net12Sparse	21	2	binary	exact	GLPF	0.00	0.00	0	1	17	8165	verified
RWMMOP15	SiteSelect	2	3	real	dense	MIX	1.00	1.00	958	2	1916	12262	verified
RWMMOP16	BlendPool	3	2	real	dense	MIX	0.90	0.90	223	3	849	1213	verified
RWMMOP17	HomotopyPath	12	2	real	homotopy	MIX	1.00	1.00	32	2	64	81	verified
RWMMOP18	ReservoirOps	8	2	real	approx	MIX	0.92	0.92	130	2	284	1847	verified
RWMMOP19	PMSched13	13	2	integer	exact	MIX	498.21	498.21	9	770	1661	102280	verified
RWMMOP20	BiTree12	24	2	binary	exact	MIX	24.51	24.51	18	41	321	2511	verified

(raw range ratio 51.65 reduced to 1.000).. *Archives:* RWMMOP1_SatelliteTTC25_full_reference.mat, RWMMOP1_SatelliteTTC25_full_metadata.txt, SatelliteTTC25_full.mat.

RWMMOP2 — Jobshop5 (job shop scheduling)

D / M / encoding:	36/2/permutation/integer
Reference tier:	high-budget-approx
Modality label:	EM
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	5.84 / 5.84†
Tie groups / max multiplicity:	163 / 8†
Front / local rows:	1041 /
Max decision separation:	1.551

Evidence: Energy-aware flexible job-shop scheduling on the frozen 5-job 11-machine instance (D=36: 18 job-label variables repaired to a legal operation sequence plus 18 machine-choice variables; M=2: idle-machine-penalised makespan and idle-machine-penalised average power (total power over makespan, both axes multiplied by unused machines + 1)); BSD-style Yarpiz licence verified from a local source archive \Paper\201907 Jobshop SSCI\CodeForSSCI2019\knee-NSGA-II\license.txt. The archive was grown from 106 to 163 rows by six independent seeds of a fixed reference search (NSGA-II and DN-NSGA-II at maxFE=20000 and N=100 plus 20000 structured random schedules per seed: 106-¿128-¿141-¿143-¿153-¿157-¿163, recorded in results\jobshop\stability_growth_phase1.log) and then declared CONVERGED when three further independent seeds neither added nor dropped a non-dominated point, so the archive is a fixed point of that search; results\jobshop\stability_curve.csv holds the verification phase (6 consecutive zero-gain seeds, appending across runs so a rerun cannot erase history) and jobshop_reference_audit fails if this evidence and the published caveat ever disagree. Equivalent multimodality is structural and verified: 4111 single-dimension perturbations of the front (every adjacent operation-label transposition and every alternative machine for every operation) were re-evaluated by the model and 878 came back bit-identical (largest rejected deviation 1596400), so each of the 163 distinct objective vectors carries equivalent schedules; multiplicity is capped at 8 per objective vector, giving a 1041-row archive with 163 exact tie groups, mean Hamming distance 3.3 of 36 positions inside a tie and separation ratio 0.93. The certificate reports EM with 2 disconnected decision-space components. Objectives are normalised with constants frozen into Jobshop5_1.mat (raw range ratio 63.16 reduced to 1.000).. *Archives:* RWMMOP2_Jobshop5_reference.mat, RWMMOP2_Jobshop5_metadata.txt, Jobshop5_1.mat.

RWMMOP3 — SatelliteTTC50 (satellite scheduling)

D / M / encoding:	352/2/integer
Reference tier:	high-budget-approx
Modality label:	EM
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	0.62 / 0.62
Tie groups / max multiplicity:	3 / 4
Front / local rows:	48 /
Max decision separation:	2.951

Evidence: case_50 scaling of the same TT&C model: 50 requests, 10 ground equipment, 463 candidate passes, 176 tasks, D=352, M=2. The class is a subclass of RWMMOP1_SatelliteTTC25 that only overrides the data hook, so the differentially validated evaluator is shared and cannot drift between the two scales. The 48-row archive is built by three-stage union sampling (uniform, feasible initialisation, constructed equivalent partners); the partners exploit a structural neutral dimension of the model - the equipment choice of an unexecuted request enters no objective - which turns equivalence multimodality into computable evidence (3 exact tie groups, largest multiplicity 4, 6 separated pairs)

rather than a sampling coincidence. Three-source objective differential is 0.0 and the objective range ratio is 1.000.. *Archives:* RWMMOP3_SatelliteTTC50_full_reference.mat, RWMMOP3_SatelliteTTC50_full_metadata.txt, SatelliteTTC50_full.mat.

RWMMOP4 — ArrayNull (antenna array pattern synthesis)

D / M / encoding:	8/2/real
Reference tier:	high-budget-approx
Modality label:	EM
Licence:	verified
MMI / MMI_{far}:	1.00 / 1.00
Tie groups / max multiplicity:	74 / 2
Front / local rows:	148 /
Max decision separation:	0.166

Evidence: Linear antenna array synthesis with prescribed nulls (D=8 real element amplitudes in [0.2, 1], M=2: peak sidelobe level in dB outside the fixed 15-degree mainlobe window, and the worst null-depth error in dB against the -60 dB target at 40 and 140 degrees; box-only, no constraints). Reversing the excitation conjugates the array factor, so both objectives are identical while the null directions map into each other - verified to 2.8e-14 and archived as copied-objective reversal partners, 74 exact tie groups of multiplicity 2 (EM). The archive CONVERGED: 22 adaptive multi-seed search seeds (NSGA-II and DN-NSGA-II at maxFE=20000, N=100, plus structured fills; 136,400 evaluations) with three consecutive zero eps-cell-gain seeds and no budget caveat. The evaluator matches an independently written scalar-loop implementation on 455 cases to 3.06e-14 relative, and the uniform excitation reproduces the textbook -12.8 dB sidelobe level. Objectives are normalised by the archive front range (range ratio 1.000).. *Archives:* RWMMOP4_ArrayNull_reference.mat, RWMMOP4_ArrayNull_metadata.txt, ArrayNull_1.mat.

RWMMOP5 — WindFarm (wind farm layout)

D / M / encoding:	16/2/real
Reference tier:	high-budget-approx
Modality label:	EM
Licence:	verified
MMI / MMI_{far}:	7.00 / 7.00
Tie groups / max multiplicity:	13 / 8
Front / local rows:	104 / 104
Max decision separation:	0.953

Evidence: Wind-farm layout optimisation on a frozen self-constructed instance (8 identical turbines, D=16 real coordinates in km, M=2: negative annual energy production in MWh under a frozen 12-sector wind rose with Jensen/Park wakes and a generic cubic power curve, and collection cable length in km as the minimum spanning tree over the turbines plus the substation). Constraints are four signed violations: convex site-polygon clearance, pairwise minimum spacing and one clearance per circular exclusion zone. The eight turbines are identical, so any permutation of the (x,y) pairs leaves both objectives unchanged - verified to 1.8e-16 relative and archived as copied-objective cyclic partners (13 exact tie groups of multiplicity 8, cap 8): the multimodality is equivalence-driven (EM). Wake shadows do make 19 further layouts locally but not globally Pareto optimal under the single-turbine 0.5 km relocation lattice (unfiltered 123-row certificate), but every one of them sits 19.7-47.2 front ranges above the front maximum on the cable axis (raw cable 9.2-18.2 km versus a front range of 2.46-2.79 km at comparable energy), so the eps=0.3 quality gate (DEC-20260928-064) removes all 19 from the published certificate, which collapses onto the 104 global rows and carries no GLPF claim; the unfiltered enumeration stays in the sidecar metadata. The archive CONVERGED: 19 adaptive multi-seed growth seeds (NSGA-II and DN-NSGA-II at maxFE=20000, N=100, plus uniform feasible fills, 17310 evaluated rows) with three consecutive zero eps-cell-gain seeds, no budget caveat.. *Archives:* RWMMOP5_WindFarm_reference.mat, RWMMOP5_WindFarm_metadata.txt, RWMMOP5_WindFarm_local_reference.mat, RWMMOP5_WindFarm_local_metadata.txt, WindFarm_1.mat.

RWMMOP6 — NestingLayout (cutting stock / nesting)

D / M / encoding:	18/2/real
Reference tier:	high-budget-approx
Modality label:	EM
Licence:	verified
MMI / MMI_{far}:	6.96 / 6.96
Tie groups / max multiplicity:	57 / 8
Front / local rows:	452 / 588
Max decision separation:	2.595

Evidence: Two-dimensional nesting (cutting-stock layout) on a frozen self-constructed instance (six parts in three identical pairs: two 2.0x1.0 rectangles, two 1.5x0.8 rectangles, two L-shaped rectangle unions; D=18 real (centre x, y and orientation per part), M=2: bounding-box area of the placed corners and negative centre spread). Constraints are signed: pairwise separating-axis penetration depths over the rectangle unions and per-part sheet-boundary excess. Rotating the whole layout by 180 degrees about the sheet centre and swapping identical part pairs preserve both objectives - verified to 2.4e-12 relative (trigonometric rounding of mod(theta+pi,2pi)) and archived as copied-objective symmetry partners (57 exact tie groups, multiplicity up to 8, cap 8); the single-part relocation lattice (+/-0.25 translation, +/-15 deg rotation) yields a published 2836-row local certificate whose 2380 surplus fixpoints are ALL front-equivalent layouts (translation slack) rather than a separated local front (0 objective-separated local-only rows; the eps=0.3 quality gate of DEC-20260928-064 removes nothing here), so the equivalent-multimodality (EM) is the operative axis. The archive CONVERGED: 26 adaptive multi-seed growth seeds (NSGA-II and DN-NSGA-II at maxFE=20000, N=100, plus uniform feasible fills and

a lattice local search to fixpoints) with three consecutive zero eps-cell-gain seeds, no budget caveat. 2026-09-30 mode filter (DEC-20260930-068): published global 456-¿452 (4 near-range symmetry duplicates), local certificate 2836-¿588 (2248 translation-slack near-duplicates pruned; published surplus is 136 far front-equivalent fixpoints, localOnly still 0); pre-filter backups in results/ref_mode_filter/pre_filter/ where the standing audit's symmetry-completeness assertion now runs.. *Archives:* RWMMOP6_NestingLayout_reference.mat, RWMMOP6_NestingLayout_metadata.txt, RWMMOP6_NestingLayout_local_reference.mat, RWMMOP6_NestingLayout_local_metadata.txt, NestingLayout_1.mat.

RWMMOP7 — EcoDispatch15MF (combined economic-emission dispatch)

D / M / encoding:	15/2/real
Reference tier:	high-budget-approx
Modality label:	EM
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	1.35 / 1.35
Tie groups / max multiplicity:	74 / 8
Front / local rows:	269 /
Max decision separation:	0.309

Evidence: Combined economic-emission dispatch with multi-fuel piecewise cost curves and valve-point loading on a frozen self-constructed 15-unit instance (D=15 real unit outputs, M=2: fuel cost per hour and emission tonnes per hour). Units (1,2), (3,4) and (5,6) are three identical twin pairs; swapping outputs inside any pair preserves both objectives exactly, and the three pairs generate a swap group of eight archived per representative (multiplicity cap 8). The piecewise fuel-switch kinks and the valve-point sinusoids make the cost surface nonconvex and multimodal. The demand equality (2600 MW) is enforced by a deterministic largest-head-room repair, so the single reported constraint is the post-repair balance residual. The archive is high-budget-approx with the consecutive zero eps-cell-gain convergence criterion (or the honest budget caveat) plus verified swap partners. 2026-09-30 mode filter (DEC-20260930-068): published set 1256-¿269 rows (987 near-range twin swaps pruned); every published equivalent pair is now separated by more than deltaFar=0.2 (MMI=MMI_far=1.346), which also closes the 2026-09-29 pending reconfirmation; pre-filter backup in results/ref_mode_filter/pre_filter/ where the standing audit's swap-group-completeness assertion now runs.. *Archives:* RWMMOP7_EcoDispatch15MF_reference.mat, RWMMOP7_EcoDispatch15MF_metadata.txt, EcoDispatch15_1.mat.

RWMMOP8 — Knapsack24 (0/1 knapsack (resource selection))

D / M / encoding:	24/2/binary
Reference tier:	exact
Modality label:	EM
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	7.76 / 7.76
Tie groups / max multiplicity:	147 / 27
Front / local rows:	924 /
Max decision separation:	3.162

Evidence: Bi-objective 0/1 knapsack with 24 items (D=24 binary, M=2: negative total value and total weight, capacity 660). Scale-up replacement of the retired Knapsack12 under DEC-20260923-061: the design space is exactly 2^24 = 16777216 subsets, exhaustively enumerated in chunks in under two seconds of vectorised time (5529292 feasible, 33.0%), so the reference stays EXACT while the front is far beyond a few thousand evaluations. Items 1-2, 3-4 and 5-6 are three identical twin pairs, so exchanging pair members leaves both objectives bit-identical: the 924 Pareto rows carry 224 distinct objective vectors in 147 exact tie groups (maximum multiplicity 27 = twin group of 8 times accidental subset-sum coincidences). No local certificate is archived: under the single-item flip neighbourhood every flip trades value against weight, so no feasible design is dominated by a neighbour and the locally Pareto set would be the entire 5.5M-row feasible set (the the retired Knapsack12 certificate was exactly this trivial set); the modality claim is therefore EM only.. *Archives:* RWMMOP8_Knapsack24_reference.mat, RWMMOP8_Knapsack24_metadata.txt, Knapsack24_1.mat.

RWMMOP9 — WTA66 (defence resource allocation)

D / M / encoding:	36/2/integer
Reference tier:	high-budget-approx
Modality label:	EM
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	6.95 / 6.95†
Tie groups / max multiplicity:	33 / 8†
Front / local rows:	258 /
Max decision separation:	2.369

Evidence: Weapon-target assignment on a frozen self-constructed 6-target, 6-weapon-type instance (D=36 integer missile allocations, M=2: negative expected retained value and ammunition cost; one fire-control constraint, at most 5 missiles per target, while inventory limits are enforced by the repair map). Scale-up replacement of the retired WTA44/the retired WTA33 under DEC-20260923-061: the inventory-feasible set is the product of the per-type composition sets, exactly 339541760652494400 allocations inside a 315^12 box, so the reference is high-budget-approx by design. The archive is the eps-front of the accumulated exact front of a multi-algorithm reference search (NSGA-II, DN-NSGA-II and CCMO, 6 seeds x 3 x 50000 evaluations

plus uniform inventory-feasible fills projected onto the salvo set): the exact front reached 1166 rows and was still growing by about 900 rows per five seeds at the stop, so row-level convergence is unreachable and the archived reference is the eps-front at 2% of the front span per axis (1.02635 and 1.34 in raw units), declared CONVERGED after 3 consecutive zero-gain seeds over the 59 occupied eps-cells. The evaluator is cross-validated against an independently written scalar-loop oracle (2264 cases, maximum relative objective difference 8.9e-17, all integer counts exact). Weapon types 3/4 and 5/6 are two pairs of twin batches (identical kill probabilities, costs and inventories), so exchanging their columns maps every plan to a plan with identical objectives: the archive stores 33 eps representatives with 225 verified twin partners (objectives copied after agreement measured, multiplicity capped at 8, the RWMMOP1/RWMMOP2 convention), giving 33 exact tie groups - EM. No local Pareto certificate exists (the feasible set cannot be enumerated), so GLPF is not claimed. Difficulty pilot (results/difficulty_gate/WTA66_pilot.csv): 6 algorithms x 3 seeds at the R2 budget, best feasible-gated IGDX 3.197, 0 of 6 algorithms mode-saturated (best median ModeCov 0.9231), best median TieDepthAll 3.09 of the archived multiplicity 8, tier=hard, accept=1.. *Archives:* RWMMOP9_WTA66_reference.mat, RWMMOP9_WTA66_metadata.txt, WTA66_1.mat.

RWMMOP10 — Truss25 (structural sizing)

D / M / encoding:	25/2/real
Reference tier:	high-budget-approx
Modality label:	EM
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	0.92 / 0.92
Tie groups / max multiplicity:	136 / 2
Front / local rows:	295 /
Max decision separation:	0.584

Evidence: Twenty-five-bar cantilever truss sizing on a frozen self-constructed five-bay symmetric instance (D=25 real member areas, M=2: total structural weight and largest free-DOF displacement; 25 signed stress violations at 250 MPa). Scale-up replacement of the retired Truss10 under DEC-20260923-061: the displacement-method evaluator is generic over the instance geometry and is cross-validated against an independently assembled finite-element oracle (tests\truss25_objective_oracle.m, results\truss25\truss25_oracle.csv: 2326 cases including all archive rows, maximum relative objective difference 3.7e-16, stress 2.5e-12, feasibility flags in full agreement). The reference is high-budget-approx: 30 independent seeds of NSGA-II and DN-NSGA-II at maxFE=50000, N=100 plus structured fills (186000 evaluated rows in total), reduced to the eps-front (0.5% of the front span per axis) because continuous fronts refine indefinitely. The eps-cell set was still gaining about one cell per five seeds at the stop (159 representatives), so the archive is budget-limited WITH a recorded caveat and the near-asymptotic check holds (+1 cell over the last five seeds) - the RWMMOP1/the retired Truss10 precedent. The instance is mirror symmetric about y=0.5 in topology and loads (verified numerically, deviation 3.4e-16), so the mirror of every front row is feasible with identical objectives: the archive stores 159 eps representatives plus 159 mirror partners with objectives copied after the agreement was measured - 159 exact tie groups, multiplicity cap 2 - EM. No local Pareto certificate exists for a 25-D continuous sizing space, so GLPF is not claimed. 2026-09-30 mode filter (DEC-20260930-068): published set 318-¿295 rows (23 near-self-symmetric mirrors pruned; MMI=MMI_far=0.922); pre-filter backup in results/ref_mode_filter/pre_filter/ where the standing audit's mirror-completeness assertion now runs. Difficulty pilot (results/difficulty_gate/Truss25_pilot.csv): 6 algorithms x 3 seeds at the R2 budget, best feasible-gated IGDX 8.35e-4, 0 of 6 algorithms mode-saturated (best median ModeCov 0.9737), best median TieDepthAll 3.585, tier=hard, accept=1.. *Archives:* RWMMOP10_Truss25_reference.mat, RWMMOP10_Truss25_metadata.txt, Truss25_1.mat.

RWMMOP11 — DNR69 (distribution network reconfiguration)

D / M / encoding:	73/3/binary
Reference tier:	exact
Modality label:	GLPF
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	0.00 / 0.00
Tie groups / max multiplicity:	0 / 1
Front / local rows:	6 / 1529
Max decision separation:	0.000

Evidence: exhaustive enumeration of all 407924 spanning trees of the frozen IEEE 69-bus feeder graph, cross-checked against the Kirchhoff matrix-tree count; 28026 feasible topologies (6.870%); 6 globally Pareto optimal; 24594 locally but not globally Pareto optimal, 1523 of which pass the eps=0.3 quality gate (DEC-20260928-064) and form the published 1529-row certificate; base loss 225.0028 kW and Vmin 0.90919 pu reproduced by the upstream solver; minimum loss 99.62 kW equals the published optimum; objectives normalised by the exact Pareto range (range ratio 1.000). *Archives:* RWMMOP11_DNR69_reference.mat, RWMMOP11_DNR69_local_reference.mat, RWMMOP11_DNR69_metadata.txt, RWMMOP11_DNR69_local_metadata.txt.

RWMMOP12 — DNR33MS (distribution network reconfiguration under peak stress)

D / M / encoding:	37/3/binary
Reference tier:	exact
Modality label:	GLPF
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	0.00 / 0.00
Tie groups / max multiplicity:	0 / 1
Front / local rows:	12 / 337
Max decision separation:	0.000

Evidence: exhaustive enumeration of all 50751 spanning trees of the frozen IEEE 33-bus feeder evaluated under two frozen load scenarios (published level x1.0 and peak stress x1.3) with the sweep solver that is differentially validated against the upstream Newton-Raphson implementation to 2.539e-12 pu; scenario-coupled feasibility (ANSI C84.1 Range B band 0.90-1.05 pu at every level, no diverged power flow) leaves 3852 of 50751 topologies (7.590%) versus 11394 at the base level alone, so the coupling is strongly binding; the initial radial configuration is feasible at base (Vmin 0.91309 pu) and infeasible at peak (Vmin 0.88392 pu), which is why reconfiguration is necessary; the Pareto front holds 12 topologies, NDSort agreeing row for row with an independent brute-force dominance filter, and its minimum base-level loss 139.5513 kW equals the published 33-bus optimum 139.55 kW; 3409 topologies are locally but not globally Pareto optimal under the single branch-exchange neighbourhood (unfiltered glpfRatio 284), of which 325 pass the eps=0.3 quality gate (DEC-20260928-064) and form the published 337-row certificate, certified by the helper that reproduces the published the retired 33-bus DNR instance certificate row for row; objectives normalised by the exact Pareto range (range ratio 1.000). *Archives:* RWMMOP12_DNR33MS_reference.mat, RWMMOP12_DNR33MS_local_reference.mat, RWMMOP12_DNR33MS_metadata.txt, RWMMOP12_DNR33MS_local_metadata.txt, DNR33MS_1.mat.

RWMMOP13 — DNR33MP (distribution network reconfiguration, two periods)

D / M / encoding:	2/3/integer
Reference tier:	exact
Modality label:	GLPF
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	0.00 / 0.00
Tie groups / max multiplicity:	0 / 1
Front / local rows:	7 / 50
Max decision separation:	0.000

Evidence: exhaustive enumeration of all 4096 pairs of candidate-pool topologies (pool of 64 frozen topologies: half by base-level loss, half by peak-level voltage deviation, from the the retired 33-bus DNR instance full enumeration); period 1 uses the published load level with the [0.93,1.05] pu window and period 2 the 1.3x peak level with the ANSI C84.1 Range B band; the exact front holds 7 designs of which 4 reuse one topology in both periods (zero switching) and 3 pay switching operations to improve a period objective, so the third objective is active; 64 designs are locally but not globally Pareto optimal under single-period reassignment, 43 of which pass the eps=0.3 quality gate (DEC-20260928-064) and form the published 50-row certificate; per-topology losses and voltage deviations come from the sweep solver differentially validated against the upstream Newton-Raphson implementation; objectives normalised by the exact Pareto range (range ratio 1.000). *Archives:* RWMMOP13_DNR33MP_reference.mat, RWMMOP13_DNR33MP_local_reference.mat, RWMMOP13_DNR33MP_metadata.txt, RWMMOP13_DNR33MP_local_metadata.txt, DNR33MP_1.mat.

RWMMOP14 — Net12Sparse (microgrid topology)

D / M / encoding:	21/2/binary
Reference tier:	exact
Modality label:	GLPF
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	0.00 / 0.00
Tie groups / max multiplicity:	0 / 1
Front / local rows:	17 / 8165
Max decision separation:	0.000

Evidence: Microgrid topology design under single-edge-failure resilience on a frozen self-constructed 12-bus instance (D=21 binary candidate-corridor selections, M=2: total line length and bridge exposure = the sum over built edges of the node pairs left unreachable when that edge is removed; one signed intact-connectivity constraint). Scale-up replacement of the retired NetworkDesign8Sparse under DEC-20260923-061: the candidate rule gives K=21, so the design space is exactly 2^21 = 2097152, exhaustively enumerated in chunks (349432 connected, 16.7%), keeping the reference EXACT while the front is far beyond a few thousand evaluations. The Warshall-closure evaluator is cross-validated against an independent union-find oracle to exact agreement (8020 cases). The 17 Pareto designs carry 17 distinct objective vectors (no exact ties), so the modality is pure GLPF: an exhaustive single-edge-flip local certificate over all connected designs finds 175994 locally Pareto designs, 175977 of them locally but not globally optimal, of which 8148 pass the eps=0.3 quality gate (DEC-20260928-064) and form the published 8165-row certificate.. *Archives:* RWMMOP14_Net12Sparse_reference.mat, RWMMOP14_Net12Sparse_metadata.txt, RWMMOP14_Net12Sparse_local_reference.mat, RWMMOP14_Net12Sparse_local_metadata.txt, Net12Sparse_1.mat.

RWMMOP15 — SiteSelect (emergency facility location)

D / M / encoding:	2/3/real
Reference tier:	certified-dense
Modality label:	MIX
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	1.00 / 1.00
Tie groups / max multiplicity:	958 / 2
Front / local rows:	1916 / 12262
Max decision separation:	0.592

Evidence: Emergency service facility location in a frozen 10 km x 6 km study area (D=2 real, M=3: distance in km to the nearest residential, commercial/industrial and transport-hub node) under eight analytic keep-out regions - a river/high-voltage corridor, a trunk-road red line, five circular exclusion zones and a convex wetland reserve - each reported as its own signed clearance, so feasibility is an exact point-location test rather than a sampled one. The two crossing corridors disconnect the feasible set into four chunks (measured by flood fill on a 0.025 km grid; 71.9% of the study area is feasible). Reference level certified-dense: a 0.005 km grid (2,403,201 nodes, 1,727,347 feasible) is reduced to the eps-front at 0.01 km... eps = 0.03 km per objective axis, one site per (f2,f3) cell that no cell to its south-west beats in f1, which is strictly non-dominated by construction and eps-dominates every feasible grid node (verified on a 20,000-point independent sample, zero violations); since each objective is 1-Lipschitz the archive is within eps + h/sqrt(2) = 0.0335 km per axis of the true front. The instance is provably mirror symmetric about x = 5 km in both its facility sets and its forbidden geography, so the 958 mirror partners added to the archive have bit-identical objectives while being up to 9 km apart in decision space: 958 exact tie groups of multiplicity 2, verified numerically to 1.33e-15 km. A relocation-lattice certificate at 0.25 km - one city block, the smallest move a planner can make - finds 10,614 locally Pareto sites of which 10,560 are locally but not globally optimal, so EM and GLPF hold at once (MIX). The evaluator matches an independently written Node.js oracle on 566 boundary-concentrated points to 8.88e-16 km, and the point-to-segment primitive matches a widely used third-party implementation on 40 random polylines and 20,641 points to 1.78e-15. 2026-09-30 mode filter (DEC-20260930-068): published global set unchanged (1916 rows); local certificate 12272-¿12262 (10 front-equivalent near-duplicate fixpoints pruned; localOnly 10346 unchanged); pre-filter backup in results/ref_mode_filter/pre_filter/. Objectives are normalised by the certified front range (range ratio 1.000).. *Archives:* RWMMOP15_SiteSelect_reference.mat, RWMMOP15_SiteSelect_local_reference.mat, RWMMOP15_SiteSelect_metadata.txt, RWMMOP15_SiteSelect_local_metadata.txt, SiteSelect_1.mat.

RWMMOP16 — BlendPool (feed-stock pooling and blending)

D / M / encoding:	3/2/real
Reference tier:	certified-dense
Modality label:	MIX
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	0.90 / 0.90
Tie groups / max multiplicity:	223 / 3
Front / local rows:	849 / 1213
Max decision separation:	1.000

Evidence: Feed-stock pooling and blending under product quality contracts (D=3 real, M=2: total feed cost and total sulphur delivered), statement revised 2026-09-30 to batch detents (DEC-20260930-068, user ruling): feeds A (tiered 6/8 from 45 units, sulphur 3) and B (13, sulphur 1) feed one pool, C (11, sulphur 1.4) bypasses it, and two identical made-to-order contracts (demand 75, sulphur spec 2.2) now ship in 25-unit batches, so the pool-to-P1 valve has the four detents 0,25,50,75 (CalDec snaps any command to the nearest detent; the evaluator applies the same snap defensively). Both objectives still depend on (uA,uB) only, so domination never depends on v1 and the problem reduces exactly to the 2-D detent-feasible footprint on the 0.25 raster (361,201 nodes, 135,814 feasible, 37.6%); every archived point owns a FINITE exact tie set - the detents inside its feasible interval - archived in full (at most 4, below the cap 8): 546 eps-front representatives plus 303 detent partners = 849 rows over 223 exact tie groups, bit-identical by construction and pairwise 25 units apart = 1/3 of the v1 range, above deltaFar=0.2 (isolated FAR equivalents; the archive is a mode-filter fixed point by construction). Reference level certified-dense: every detent-feasible raster node is verified eps-dominated at eps=(1.0,0.2) per axis, and with Lipschitz constants sqrt(29), sqrt(2) the archive is within 1.952 and 0.4915 units per axis of the true front. The detent gaps create genuine band-edge traps: a 0.125 relocation lattice with a 0.5 Euclidean neighbourhood certifies 364 locally-but-not-globally Pareto plans, all passing the eps=0.3 quality gate, so the published 1213-row certificate carries GLPF content and the label is MIX. The evaluator matches an independently written loop-based oracle on 20,008 random draws plus all 849 archive rows to 4.55e-13 with full feasibility agreement, and detent neutrality is verified bit-exactly on 126 multi-detent tie intervals (widest 74.93 units). Difficulty pilot tier=hard, accept=1 (best feasible-gated IGDX 2.038, 1 of 6 algorithms mode-saturated). The pre-revision continuous-v1 statement (near-range neutral chain, verdict near_only, DEC-20260929-067) was frozen in archive/c31_redesign_2026-09-30/ (archive deleted in the 2026-09-30 cleanup round, DEC-20260930-069). Objectives are normalised by the certified front range (range ratio 1.000).. *Archives:* RWMMOP16_BlendPool_reference.mat, RWMMOP16_BlendPool_local_reference.mat, RWMMOP16_BlendPool_metadata.txt, RWMMOP16_BlendPool_local_metadata.txt, BlendPool_1.mat.

RWMMOP17 — HomotopyPath (robot path planning)

D / M / encoding:	12/2/real
Reference tier:	homotopy-certified
Modality label:	MIX
Licence:	verified
MMI / MMI_{far}:	1.00 / 1.00
Tie groups / max multiplicity:	32 / 2
Front / local rows:	64 / 81
Max decision separation:	1.830

Evidence: Continuous waypoint path planning through a frozen obstacle field (K=6 waypoints, D=12 real, M=2: polyline length in km and risk exposure as the sum over obstacles and segments of max(0, (r+1.5) - point-segment distance)ˆ2). Constraints are four signed obstacle clearances. Mirroring y -¿ 7 - y preserves both endpoints, the obstacle set and both objectives - verified to 3e-16 relative and archived as copied-objective mirror partners (32 exact tie groups of multiplicity 2, cap 8); the homotopy classes at the two axis circles are separated by infeasible crossings and 17 further layouts are locally but not globally Pareto optimal under the 0.25 km single-waypoint relocation lattice (81-row local certificate; all 17 pass the eps=0.3 quality gate of DEC-20260928-064 unchanged), so EM and GLPF hold at once (MIX). Reference level homotopy-certified: the four side-sign classes are enumerated explicitly, each class front grown by a class-restricted structured search (3 of 4 classes converged with three consecutive zero eps-cell-gain seeds, one budget-limited with caveat), the archive is the nondominated union after lattice-domination screening (64 rows).. *Archives:* RWMMOP17_HomotopyPath_reference.mat, RWMMOP17_HomotopyPath_metadata.txt, RWMMOP17_HomotopyPath_local_reference.mat, RWMMOP17_HomotopyPath_local_metadata.txt, HomotopyPath_1.mat.

RWMMOP18 — ReservoirOps (reservoir operation)

D / M / encoding:	8/2/real
Reference tier:	high-budget-approx
Modality label:	MIX
Licence:	verified
MMI / MMI_{far}:	0.92 / 0.92
Tie groups / max multiplicity:	130 / 2
Front / local rows:	284 / 1847
Max decision separation:	1.622

Evidence: Parallel-reservoir flood-control vs hydropower operation on a frozen self-constructed instance (two identical reservoirs, four periods, D=8 real releases in [0,250], M=2: downstream flood peak = max over periods of total release plus lateral inflow, and negative hydropower = -sum head*release with head from the storage trajectory). Constraints are four signed violations (per-reservoir worst storage bound violation and per-reservoir terminal-target violation). The two reservoirs are identical, so swapping their release trajectories preserves both objectives bit-exactly (exact EM, multiplicity 2 per representative); distinct operation policies (pre-release vs hold vs late release) are local Pareto regions under the per-release relocation lattice (GLPF: 1885 local-only fixpoints enumerated, 1563 of which pass the eps=0.3 quality gate of DEC-20260928-064 and form the published 1871-row certificate). The archive is high-budget-approx with the consecutive zero eps-cell-gain convergence criterion plus verified swap partners; the local certificate is the lattice fixpoint set union the archived global rows. 2026-09-30 mode filter (DEC-20260930-068): published global 308-¿284 (24 near-range swaps pruned), local certificate 1871-¿1847 (24 near-duplicates pruned; localOnly 1563 unchanged); pre-filter backups in results/ref_mode_filter/pre_filter/ where the standing audit's swap-completeness assertion now runs.. *Archives:* RWMMOP18_ReservoirOps_reference.mat, RWMMOP18_ReservoirOps_metadata.txt, RWMMOP18_ReservoirOps_local_reference.mat, RWMMOP18_ReservoirOps_local_metadata.txt, ReservoirOps_1.mat.

RWMMOP19 — PMSched13 (parallel machine scheduling)

D / M / encoding:	13/2/integer
Reference tier:	exact
Modality label:	MIX
Licence:	verified
MMI / MMI_{far}:	498.21 / 498.21
Tie groups / max multiplicity:	9 / 770
Front / local rows:	1661 / 102280
Max decision separation:	1.202

Evidence: Identical parallel machine scheduling with 13 jobs on 3 machines (D=13 integer machine indices, M=2: makespan and total processing cost; job 13 ineligible on machine 1). Scale-up replacement of the retired PMSched8 under DEC-20260923-061: the design space is exactly 3ˆ13 = 1594323 assignments, exhaustively enumerated in chunks (1062882 feasible, 66.7%), so the reference stays EXACT while the front is no longer reachable by a few thousand evaluations. Machines 2 and 3 are identical columns, so exchanging any subset of jobs between them leaves both objectives bit-identical: the 1661 Pareto rows carry only 13 distinct objective vectors in 9 exact-tie groups (maximum multiplicity 770). The exhaustive single-job-move local certificate over every feasible design finds 103473 locally Pareto designs, 101812 of them locally but not globally optimal, of which 100619 pass the eps=0.3 quality gate (DEC-20260928-064) and form the published 102280-row certificate (strong GLPF content), so EM and GLPF hold at once (MIX).. *Archives:* RWMMOP19_PMSched13_reference.mat, RWMMOP19_PMSched13_metadata.txt, RWMMOP19_PMSched13_local_reference.mat, RWMMOP19_PMSched13_local_metadata.txt, PMSched13_1.mat.

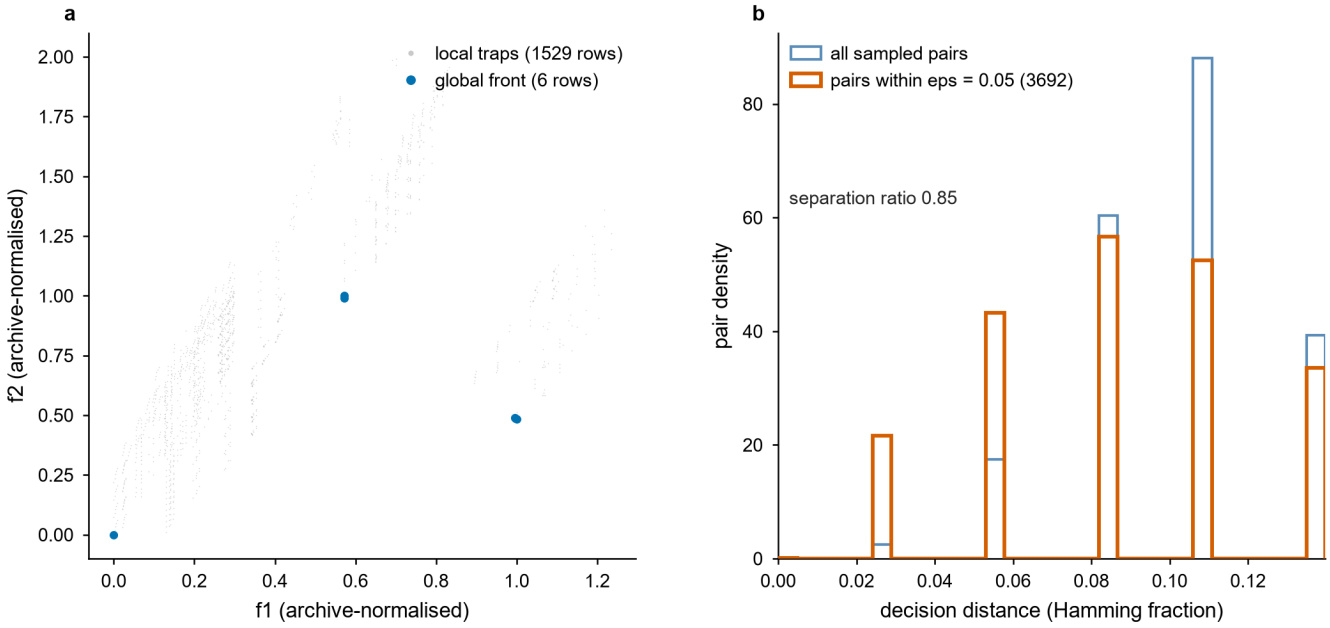
RWMMOP20 — BiTree12 (network design (spanning tree))

D / M / encoding:	24/2/binary
Reference tier:	exact
Modality label:	MIX
Licence:	verified
MMI / $\mathrm{MMI}_{\mathrm{far}}$:	24.51 / 24.51
Tie groups / max multiplicity:	18 / 41
Front / local rows:	321 / 2511
Max decision separation:	3.162

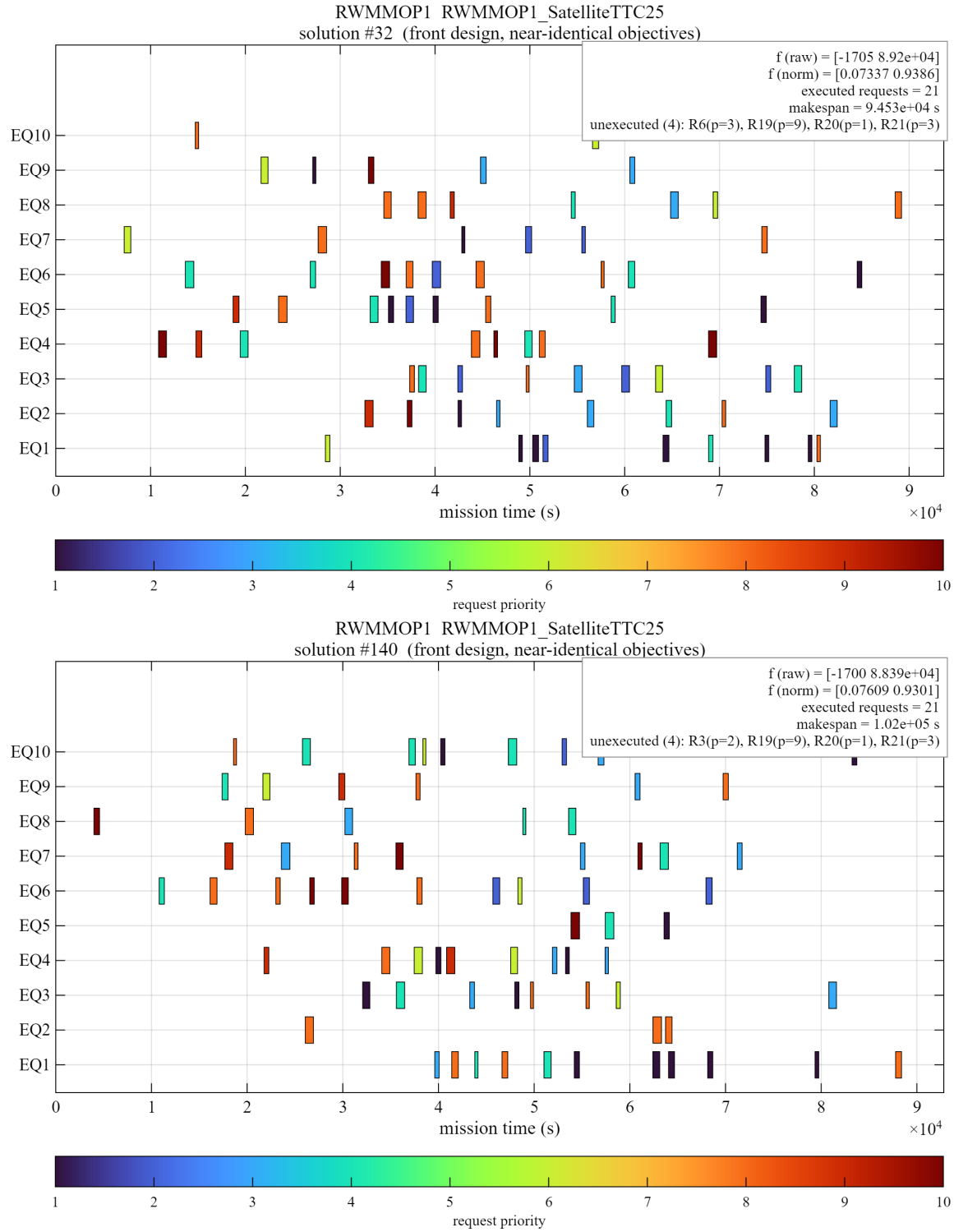
Evidence: Bi-criterion degree-bounded spanning tree on a frozen 12-node 24-edge graph (cycle plus twelve chords; D=24 binary edge selections, M=2: construction cost and maintenance cost; one signed constraint: (1-isTree)+degree violation, so the feasible set is exactly the 272010 bounded-degree spanning trees). Scale-up replacement of the retired BiTree8 under DEC-20260923-061: the size-11 subset space is C(24,11) = 2496144, exhaustively enumerated with vectorised degree/connectivity filters in about 23 seconds, so the reference stays EXACT while the feasible set is far beyond a few thousand evaluations. Two chord pairs carry identical (wA,wB) weights: exchanging a twin chord inside a tree keeps both objectives bit-identical (321 Pareto rows over only 19 distinct objective vectors). The exhaustive single-edge-exchange local certificate over all 272010 trees finds 2511 locally Pareto trees, 2190 of them locally but not globally optimal (GLPF), so EM and GLPF hold at once (MIX).. *Archives:* RWMMOP20_BiTree12_reference.mat, RWMMOP20_BiTree12_metadata.txt, RWMMOP20_BiTree12_local_reference.mat, RWMMOP20_BiTree12_local_metadata.txt, BiTree12_1.mat.

S2. Equivalent-Design Showcase

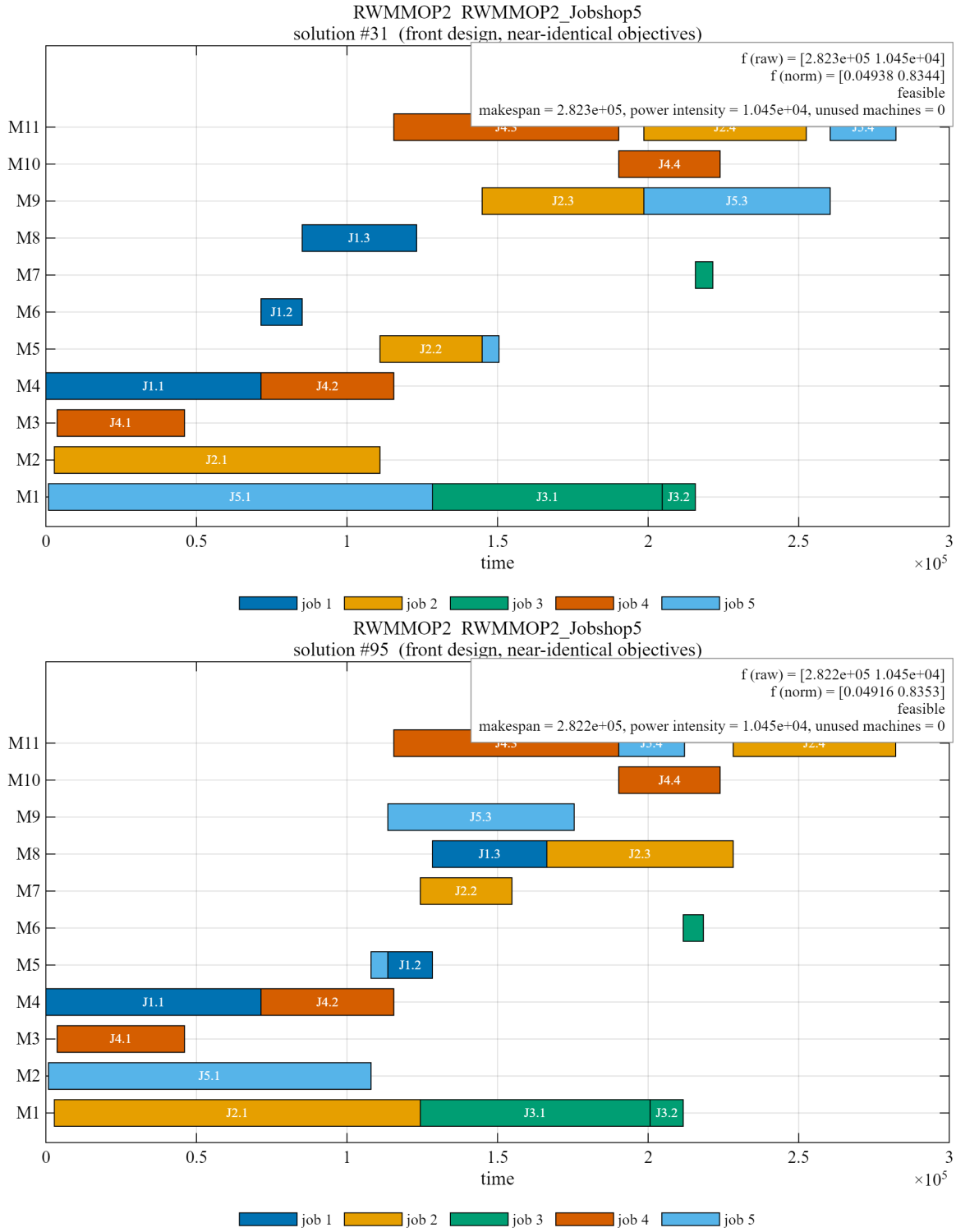
Each block below places two optimal solutions of one problem side by side (design A on top, design B below). The pair is chosen by a deterministic, render-checked rule: among the published archive's closest-objective design pairs, the one whose two domain renders differ most. For twelve problems that pair sits inside an exact tie group (objectives bit-identical, designs materially different). For RWMMOP1, RWMMOP2, RWMMOP3 and RWMMOP5 every exact tie renders as one identical picture — the equivalence is a pure relabelling or permutation of identical units — so the pair is instead the nearest-objective visibly distinct pair of the global front. For the four GLPF problems (RWMMOP11–14), whose global fronts are realised by one connected decision region each (MMI $= 0$ by complete enumeration), no equivalent global optima exist; the pair is instead the trap witness — one global optimum and the certified local optimum nearest to it in objective space. Every render is produced by the problem's own domain renderer (the same code the interactive GUI uses) from the frozen archives; the pair selection, the objectives, the decision distances and the captions are regenerated and re-validated by the project's regression suite (`rwmmo_render_si_showcase`, `write_si_showcase`, `validate_si_showcase`); no number is typed by hand. Decision distances are archive-range-normalised Euclidean distances; the objective gap Δf is the Euclidean distance in the archive-normalised objective space.



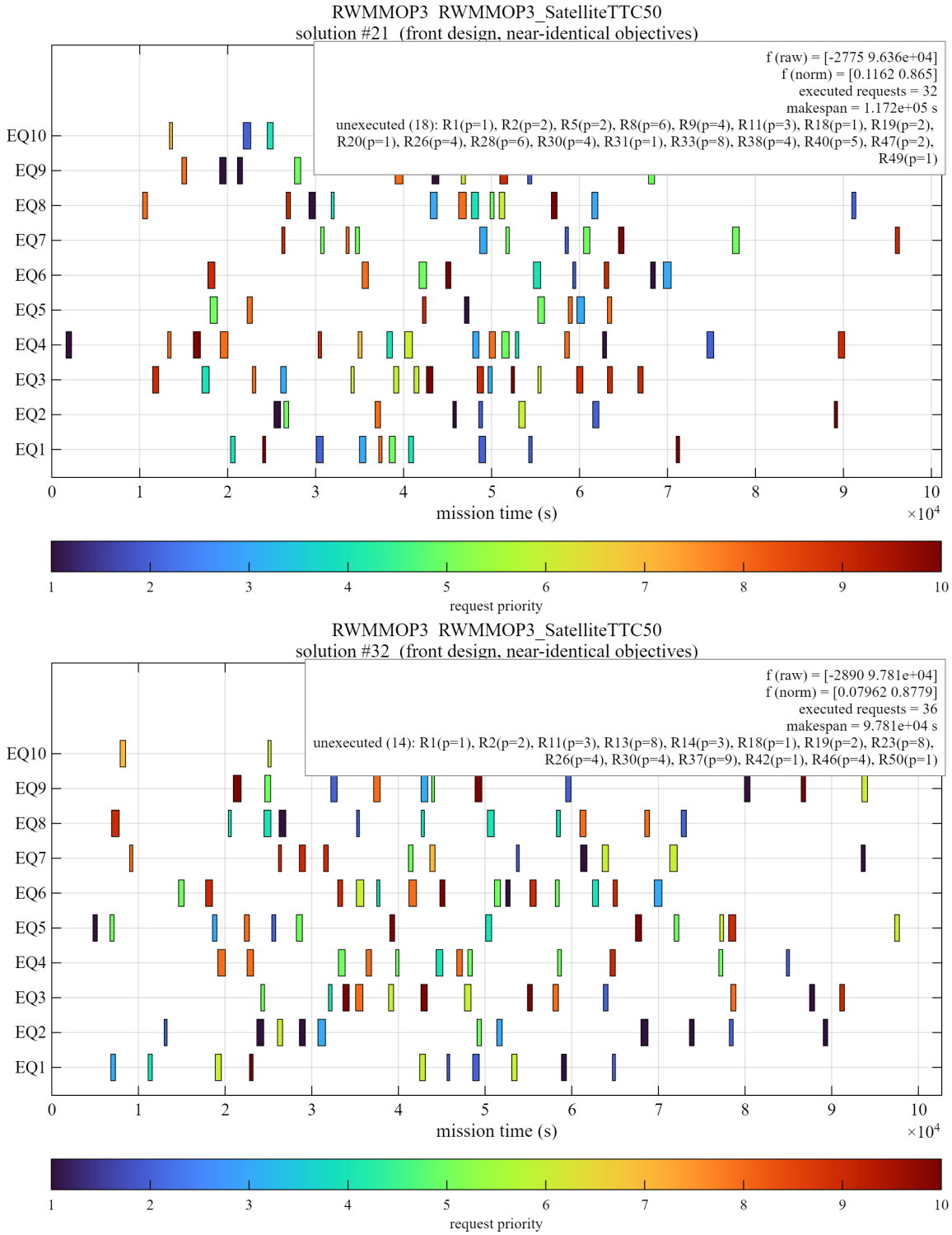
Supplementary Fig. S4: Trap-axis witness RWMMOP11 (DNR69). a, The six-row exact global front (complete enumeration of 407,924 spanning trees) in front of the 1529-row published local-optimum certificate (objectives 1–2 of 3, archive-normalised). b, Distribution of decision distances (Hamming fraction) over all pairs of a fixed 600-row sample of front plus certificate (blue outline) and over the pairs whose objectives agree within $\varepsilon = 0.05$ (vermillion): the two coincide (certificate separation ratio 0.85), i.e. near-objective pairs stay decision-distant, the trap-axis counterpart of equivalent-PS separation.



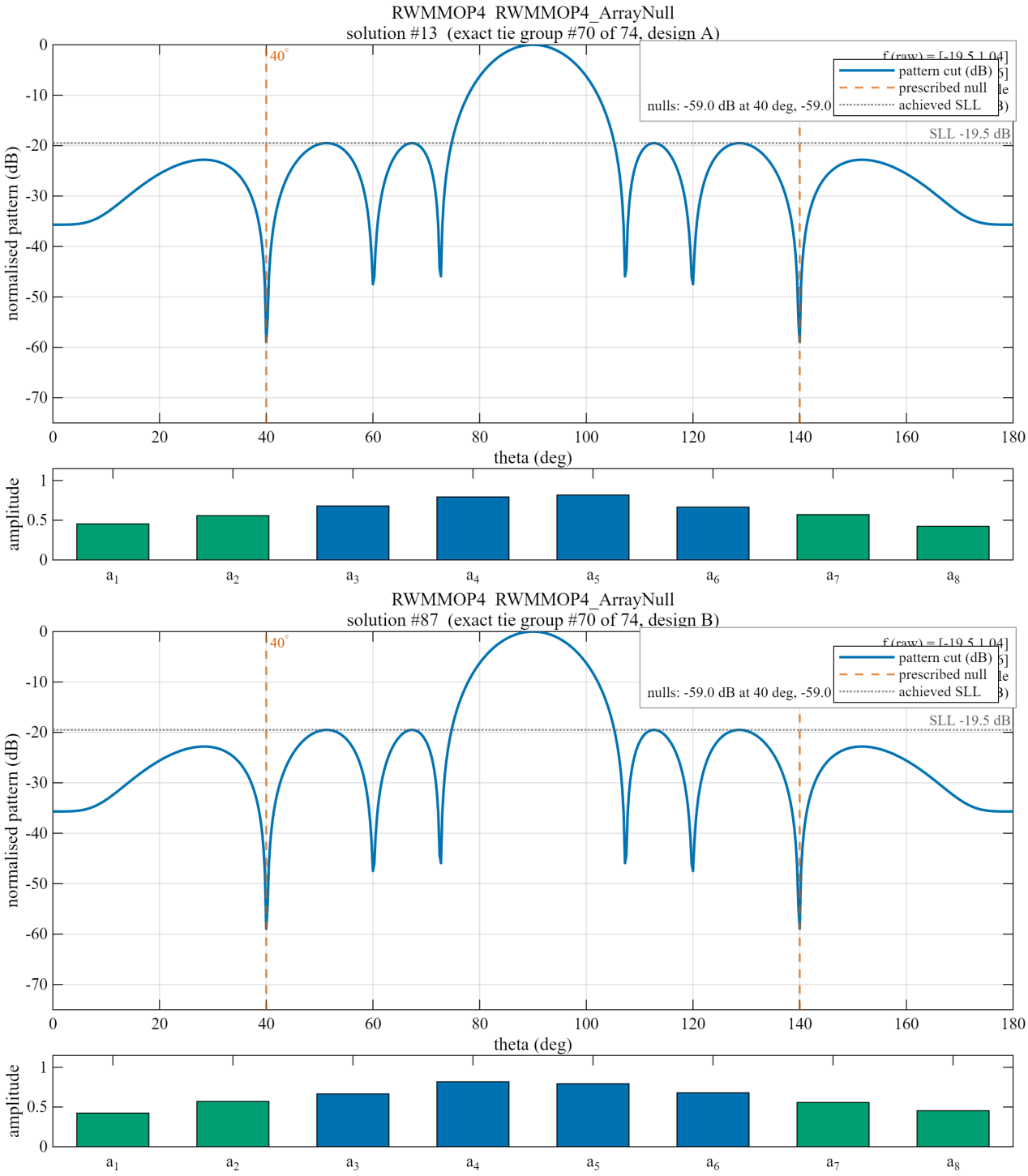
RWMMOP1 **SatelliteTTC25** (EM): designs A (top) and B (bottom) are the visually most different pair among the closest-objective design pairs of the front (global rows 32 and 140, $\Delta f = 0.00887$, archive-normalised). Yet 122 of the 164 decision coordinates differ (decision distance 5.39). Two full schedules executing 21 of the 25 requests at near-identical reward and finish time: 122 of the 164 request slots carry different equipment or lap choices.



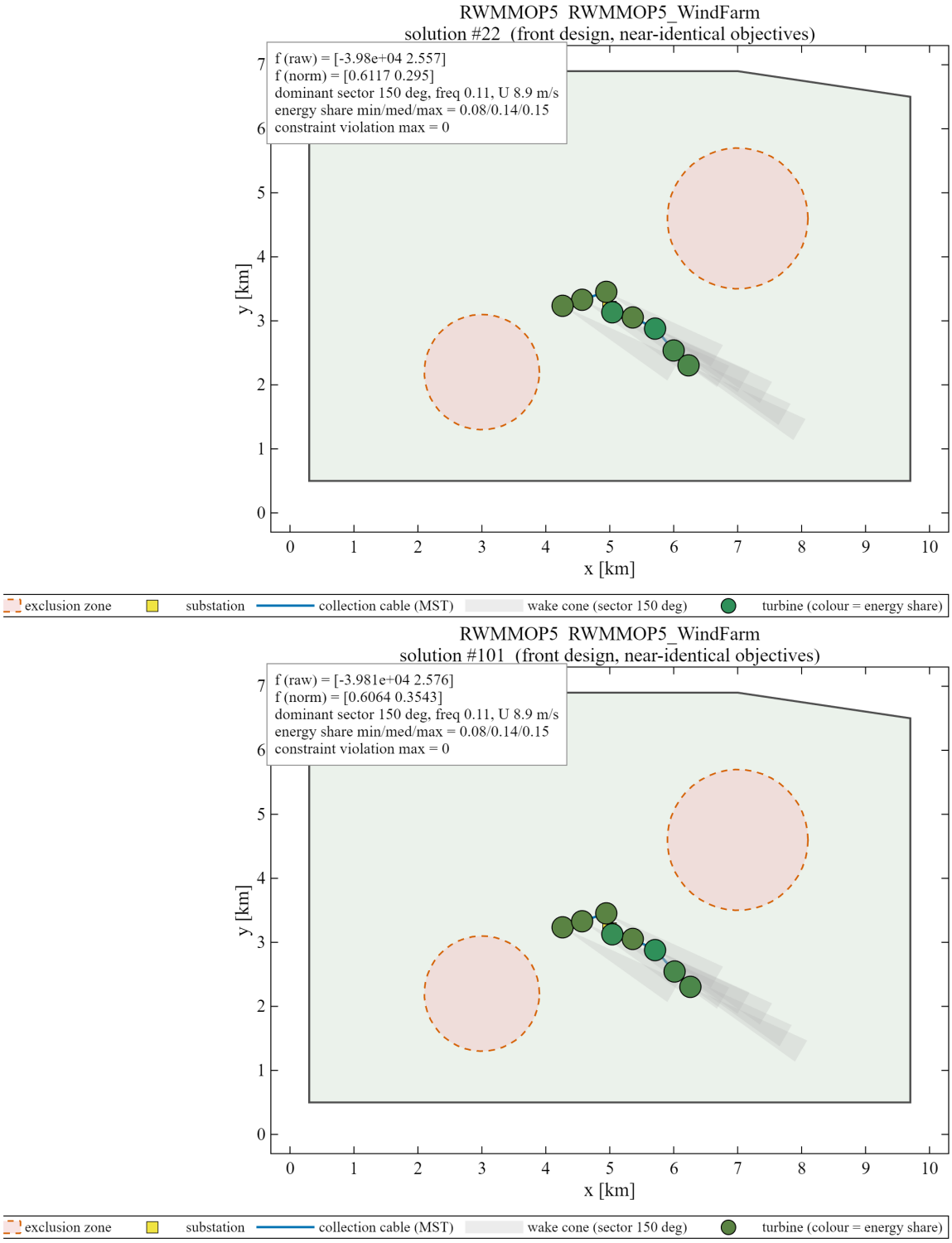
RWMMOP2 **Jobshop5** (EM): designs A (top) and B (bottom) are the visually most different pair among the closest-objective design pairs of the front (global rows 31 and 95, $\Delta f = 0.00095$, archive-normalised). Yet 23 of the 36 decision coordinates differ (decision distance 2.77). The two schedules differ in 23 of the 36 sequence and machine slots, yet the makespan moves by one part in 280 and the power intensity not at all.



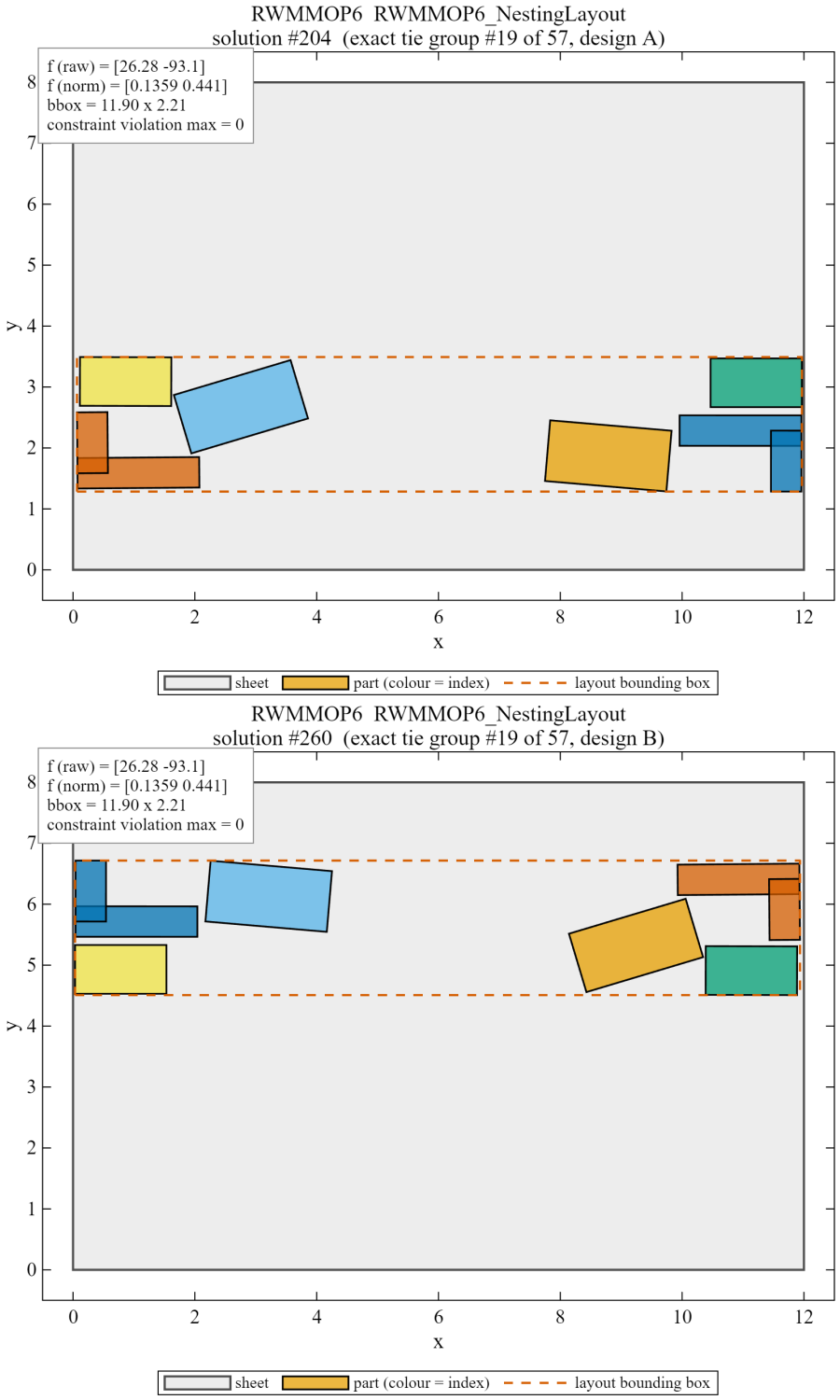
RWMMOP3 **SatelliteTTC50** (EM): designs A (top) and B (bottom) are the visually most different pair among the closest-objective design pairs of the front (global rows 21 and 32, $\Delta f = 0.0389$, archive-normalised). Yet 270 of the 352 decision coordinates differ (decision distance 7.49). 270 of the 352 slots change — almost every request carries different equipment and lap choices — while reward and finish time stay within a 0.039 normalised gap.



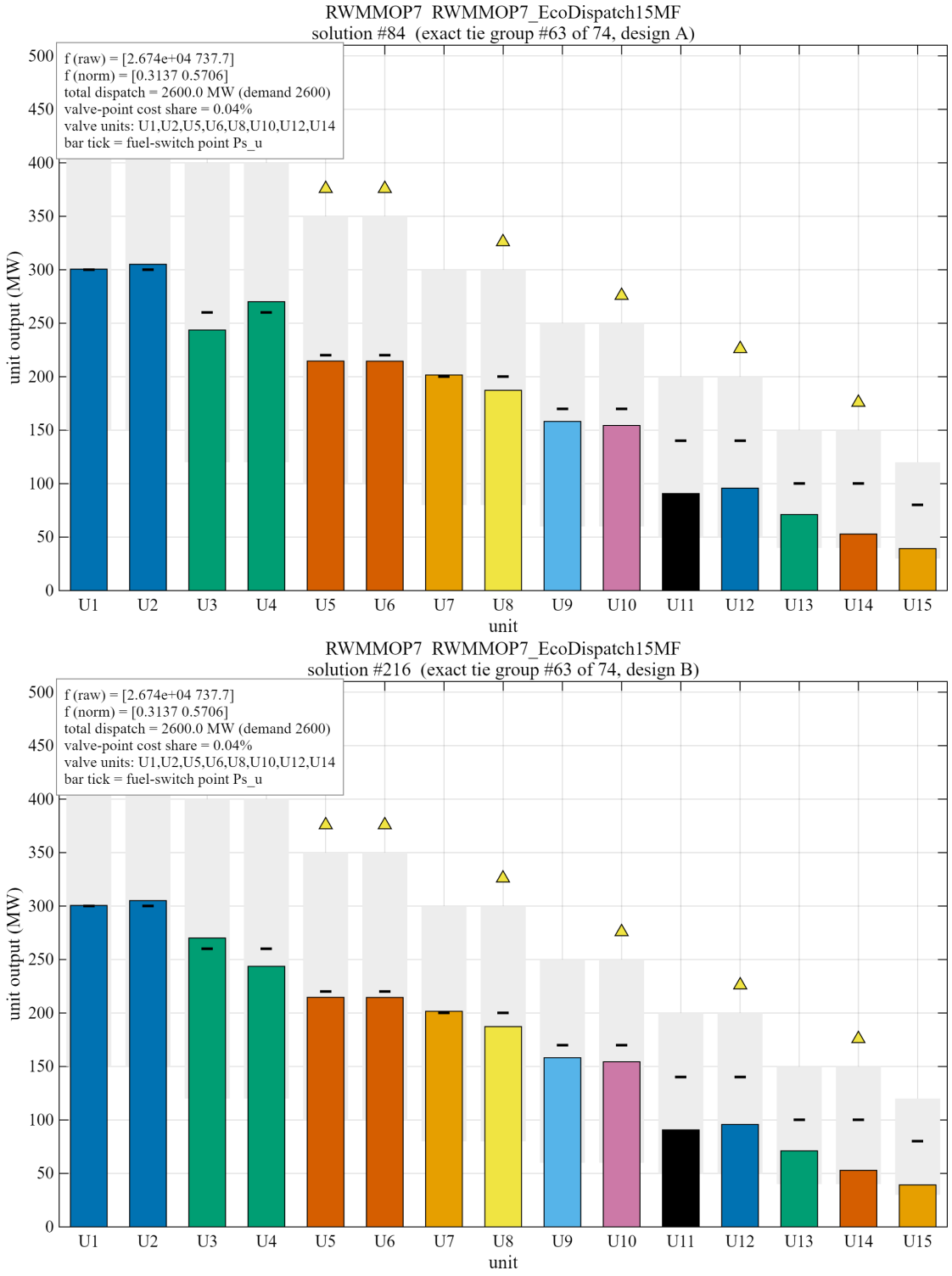
RWMMOP4 **ArrayNull** (EM): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #70 (2 members in the published archive, global rows 13 and 87). Objectives bit-identical, $(f_1, f_2) = (-19.4971,\ 1.03999)$; yet all 8 decision coordinates ($\{1, 2, 3, 4, 5, 6, 7, 8\}$) differ (decision distance 0.453). All eight element excitations take different values, yet the pattern keeps the same null depth and the same peak sidelobe level.



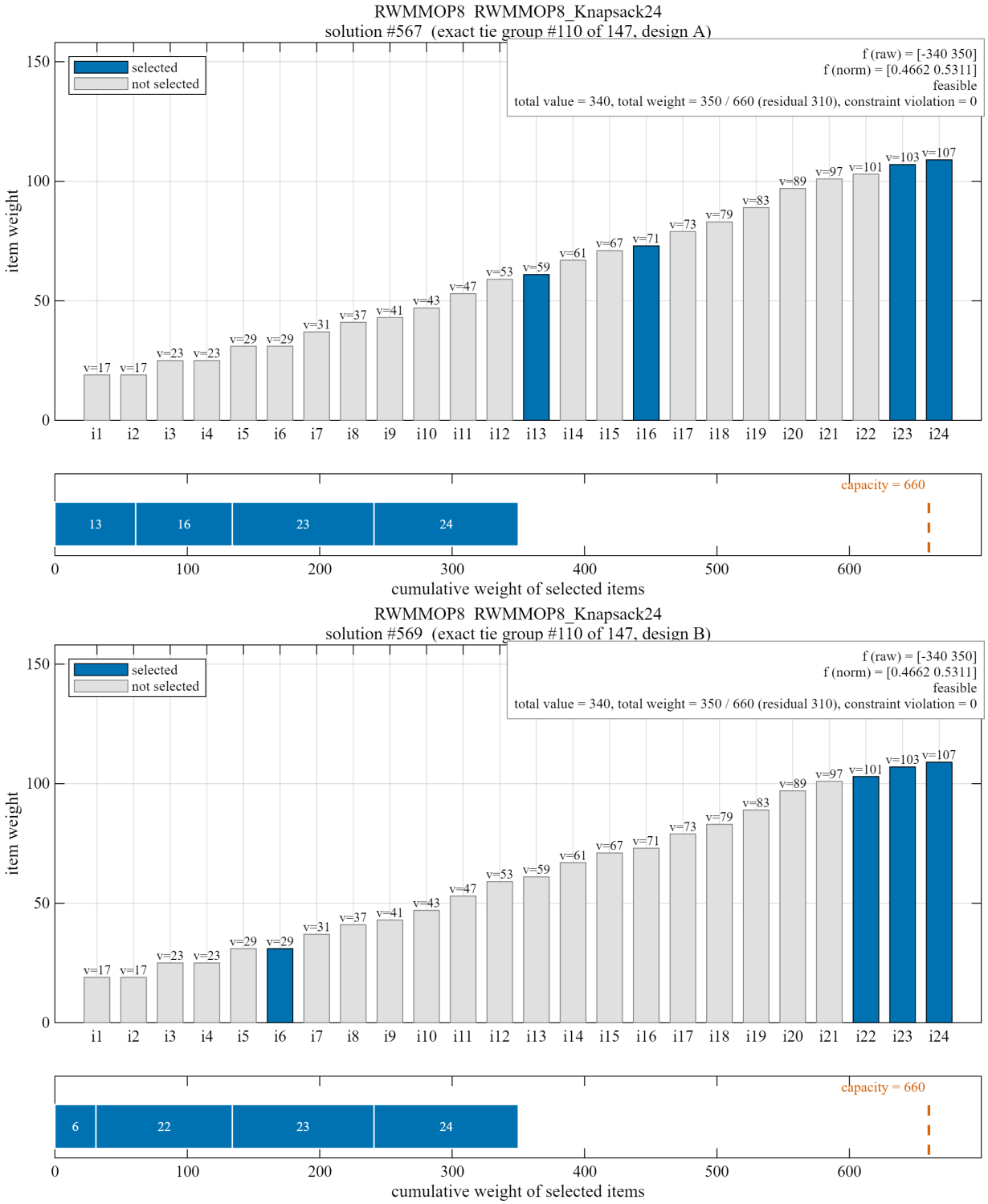
RWMMOP5 **WindFarm** (EM): designs A (top) and B (bottom) are the visually most different pair among the closest-objective design pairs of the front (global rows 22 and 101, $\Delta f = 0.0595$, archive-normalised). Yet all 16 decision coordinates differ (decision distance 1.25). Every one of the eight turbines stands at a different coordinate; the two layouts' objective values differ by only 0.06 on the normalised front scale.



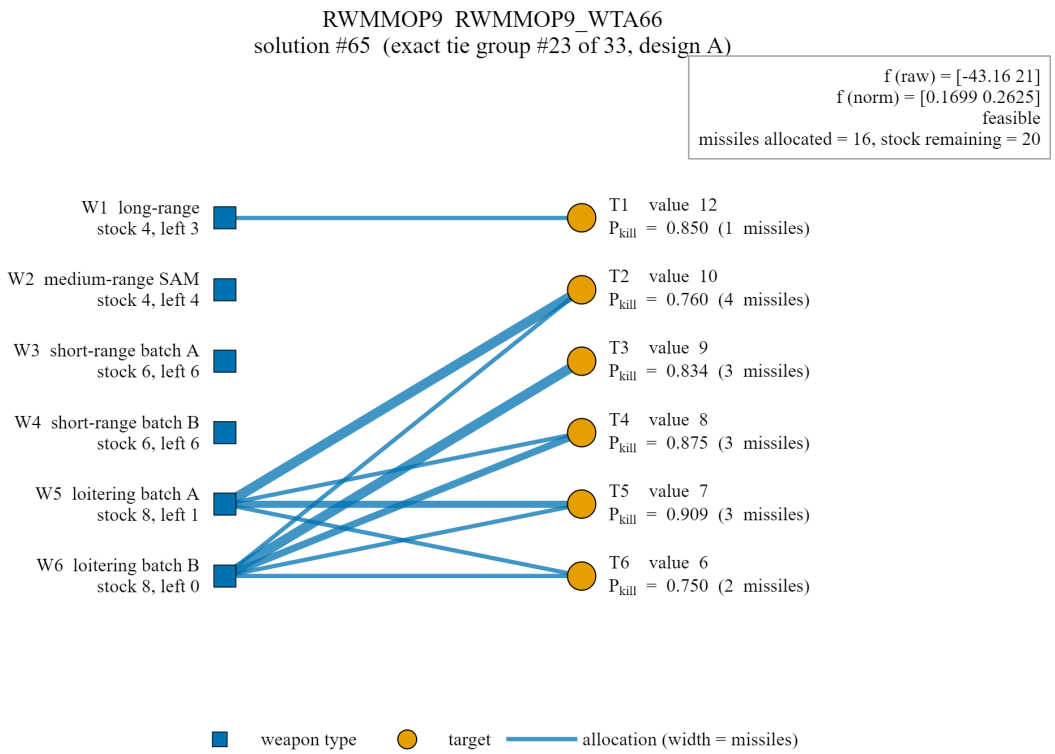
RWMMOP6 **NestingLayout** (EM): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #19 (8 members in the published archive, global rows 204 and 260). Objectives bit-identical, $(f_1, f_2) = (26.2755, -93.0988)$; yet all 18 decision coordinates differ (decision distance 2.33). Every piece sits at a different position (all 18 placement coordinates change); the two nesting layouts return identical objective values.

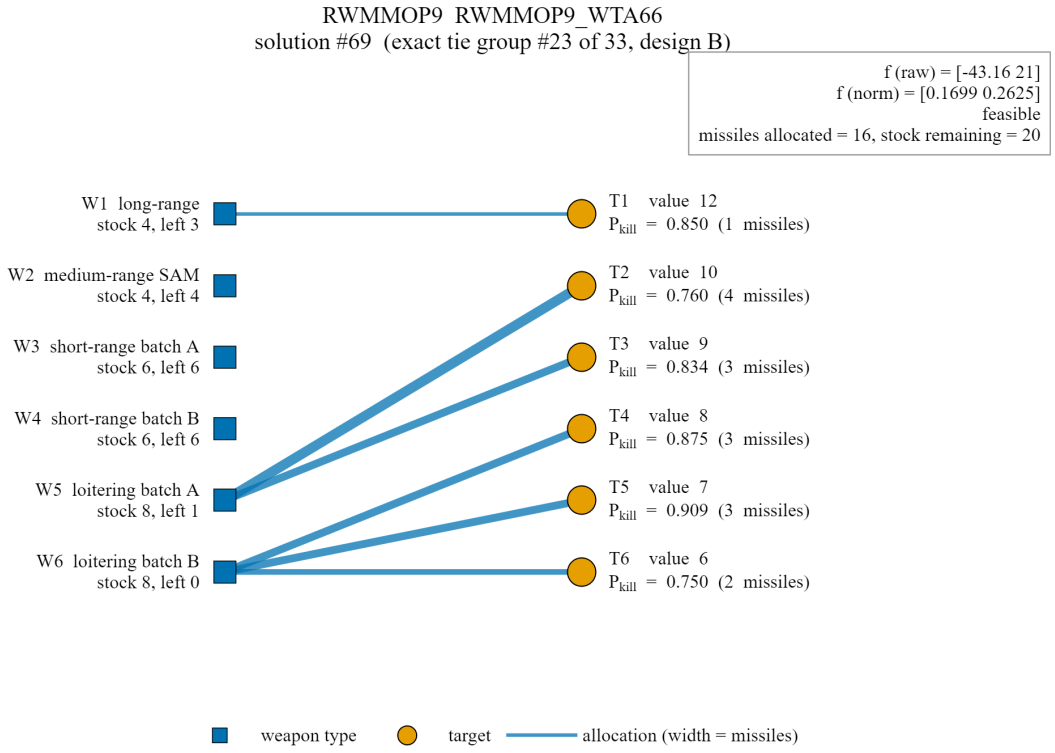


RWMMOP7 **EcoDispatch15MF** (EM): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #63 (2 members in the published archive, global rows 84 and 216). Objectives bit-identical, $(f_1, f_2) = (26736.2,\ 737.692)$; yet 2 of the 15 decision coordinates ($\{3, 4\}$) differ (decision distance 0.219). Only the outputs of units 3 and 4 change; total fuel cost and total emission stay bit-identical.

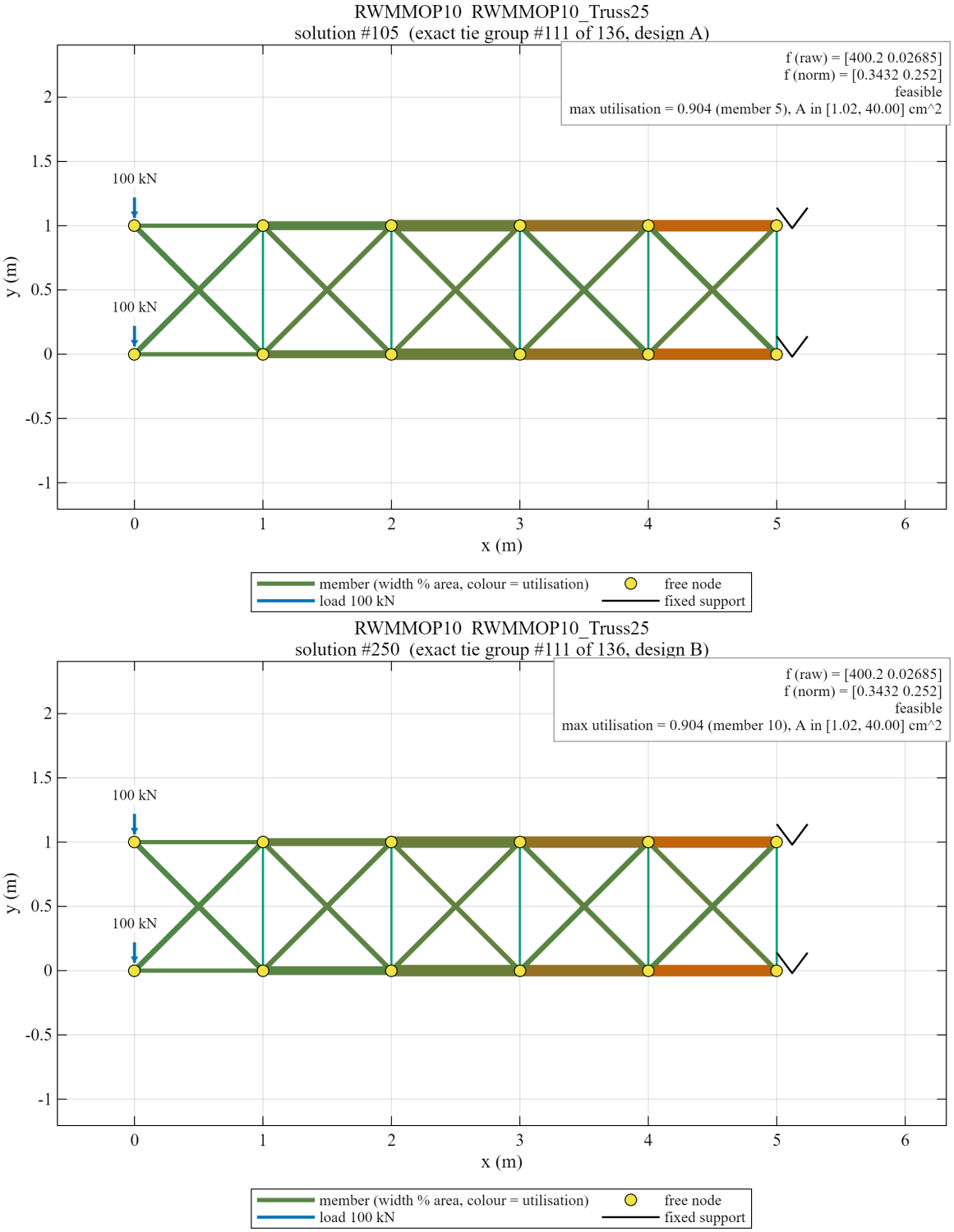


RWMMOP8 **Knapsack24** (EM): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #110 (3 members in the published archive, global rows 567 and 569). Objectives bit-identical, $(f_1, f_2) = (-340,\ 350)$; yet 4 of the 24 decision coordinates ($\{6, 13, 16, 22\}$) differ (decision distance 2). Four items (i6, i13, i16 and i22) trade membership; the two packs weigh and score exactly the same.

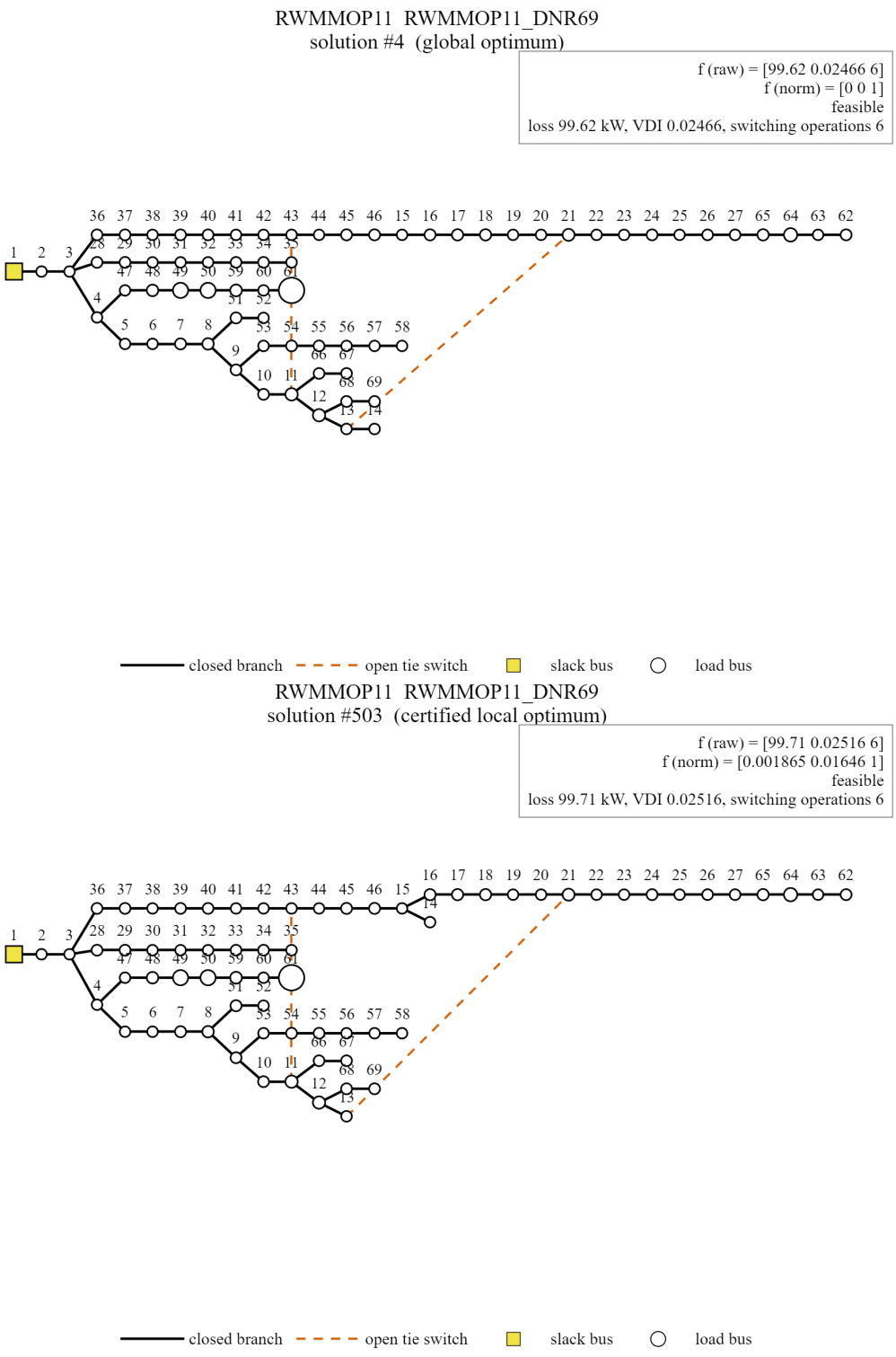




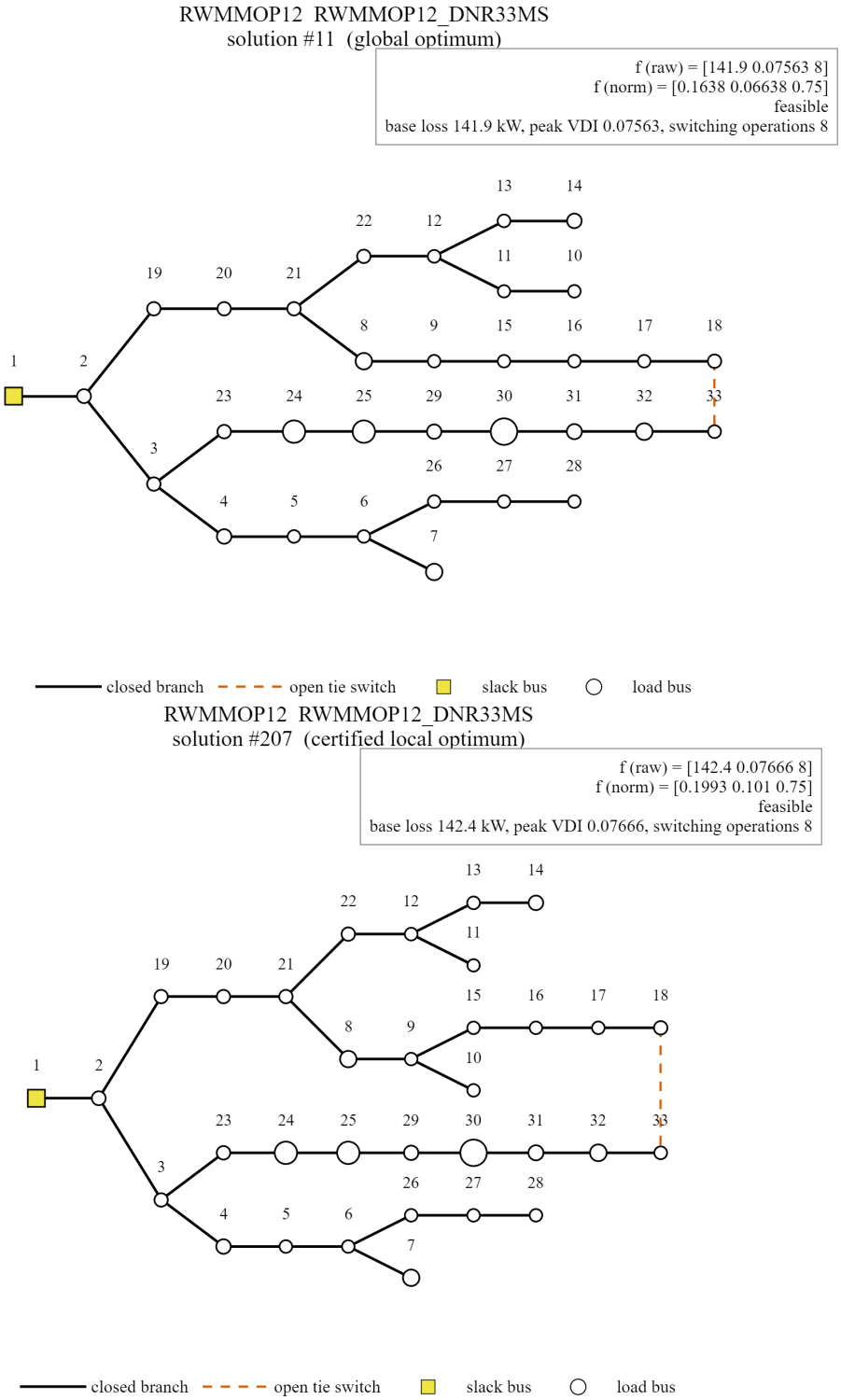
RWMMOP9 **WTA66** (EM): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #23 (8 members in the published archive, global rows 65 and 69). Objectives bit-identical, $(f_1, f_2) = (-43.1638,\ 21)$; yet 10 of the 36 decision coordinates ($\{26, 27, 28, 29, 30, 32, 33, 34, 35, 36\}$) differ (decision distance 1.75). Ten engagements (positions 26–30 and 32–36) are retargeted; expected damage and assignment cost coincide to the last bit.



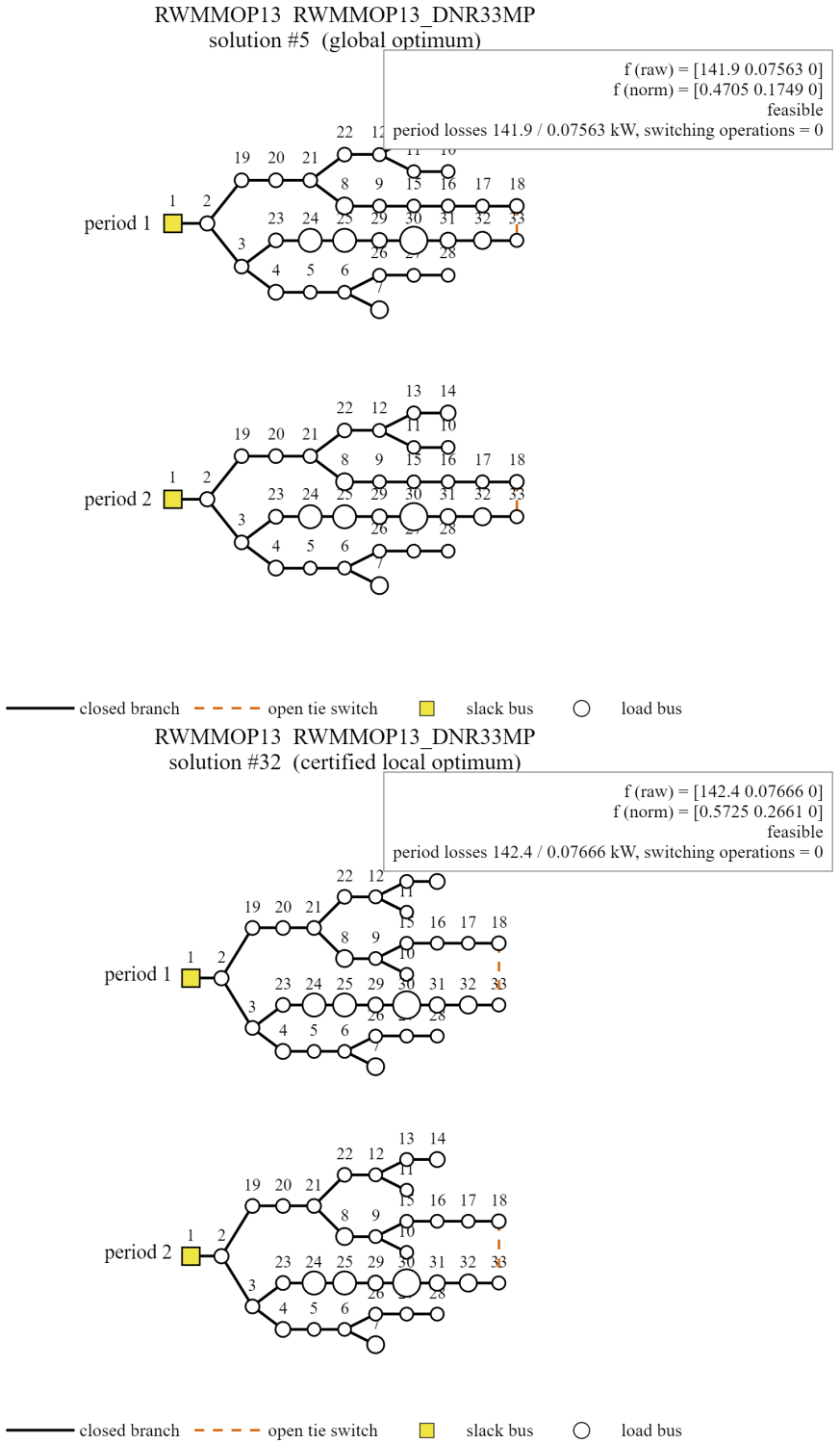
RWMMOP10 **Truss25** (EM): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #111 (2 members in the published archive, global rows 105 and 250). Objectives bit-identical, $(f_1, f_2) = (400.236,\ 0.0268457)$; yet 20 of the 25 decision coordinates differ (decision distance 0.216). Twenty of the 25 member cross-sections change — the mirror-symmetric sizing of the truss — at identical mass and identical compliance.



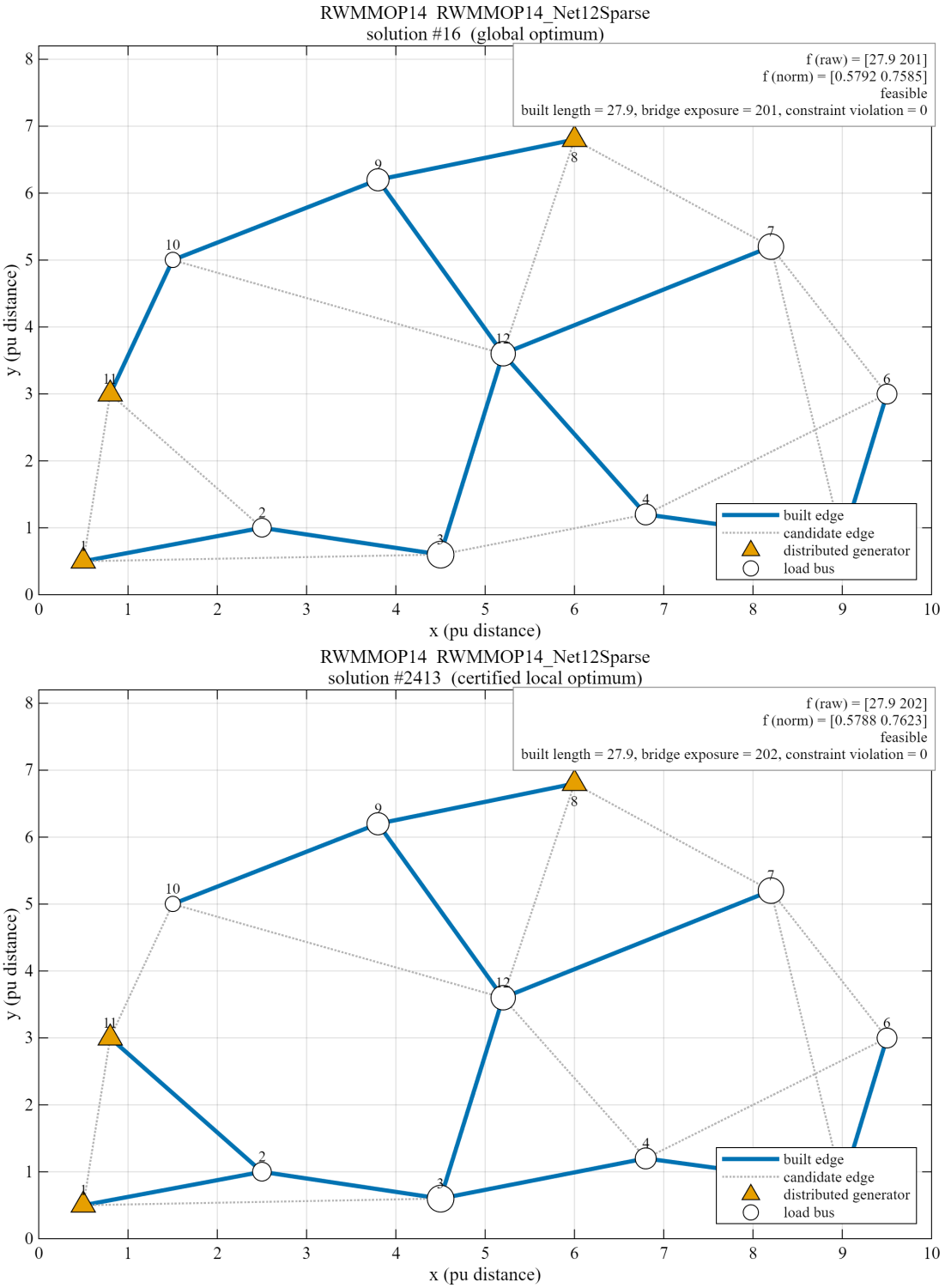
RWMMOP11 **DNR69** (GLPF): A (top) is a global optimum (front row 4); B (bottom) is a certified local optimum (certificate row 503) — the most decision-distant certificate row within twice the nearest-objective gap, objectives only $\Delta f = 0.0166$ from A's: $(f_1, f_2, f_3) = (99.6203,\ 0.0246566,\ 6)$ vs $(99.7146,\ 0.0251618,\ 6)$. Yet 2 of the 73 decision coordinates ($\{13, 14\}$) differ (decision distance 1.41). The certified local optimum relocates one open switch (branch states 13 and 14 flip); its loss is only 0.094 kW higher (99.71 vs 99.62 kW) at equal switching operations — it looks competitive but is dominated.



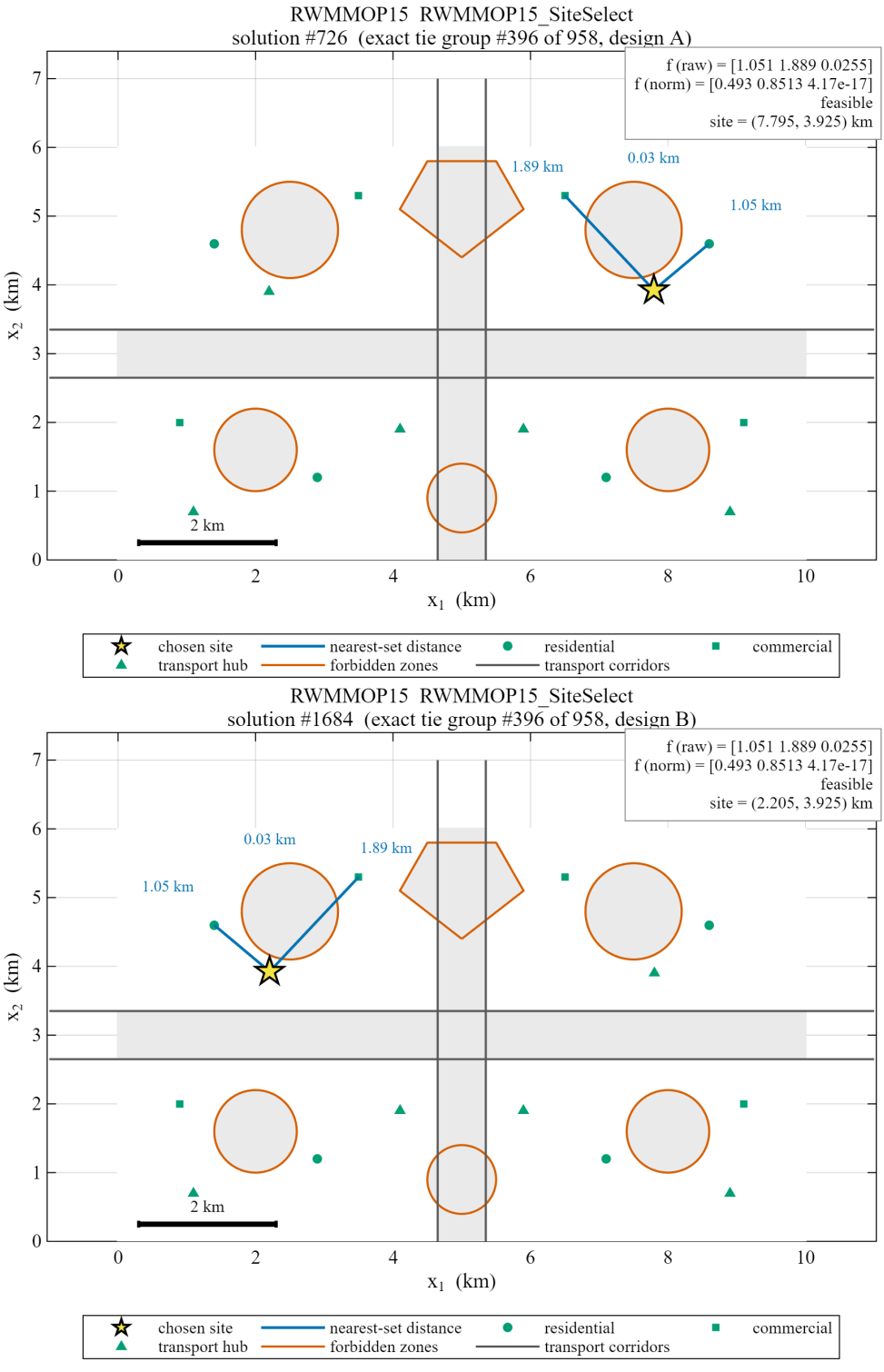
RWMMOP12 **DNR33MS** (GLPF): A (top) is a global optimum (front row 11); B (bottom) is a certified local optimum (certificate row 207) — the most decision-distant certificate row within twice the nearest-objective gap, objectives only $\Delta f = 0.0496$ from A's: $(f_1, f_2, f_3) = (141.916,\ 0.0756302,\ 8)$ vs $(142.429,\ 0.0766583,\ 8)$. Yet 2 of the 37 decision coordinates ($\{9, 10\}$) differ (decision distance 1.41). One open switch moves (branch states 9 and 10 flip); the certified local optimum loses only 0.51 kW (142.43 vs 141.92 kW) at equal switching operations.



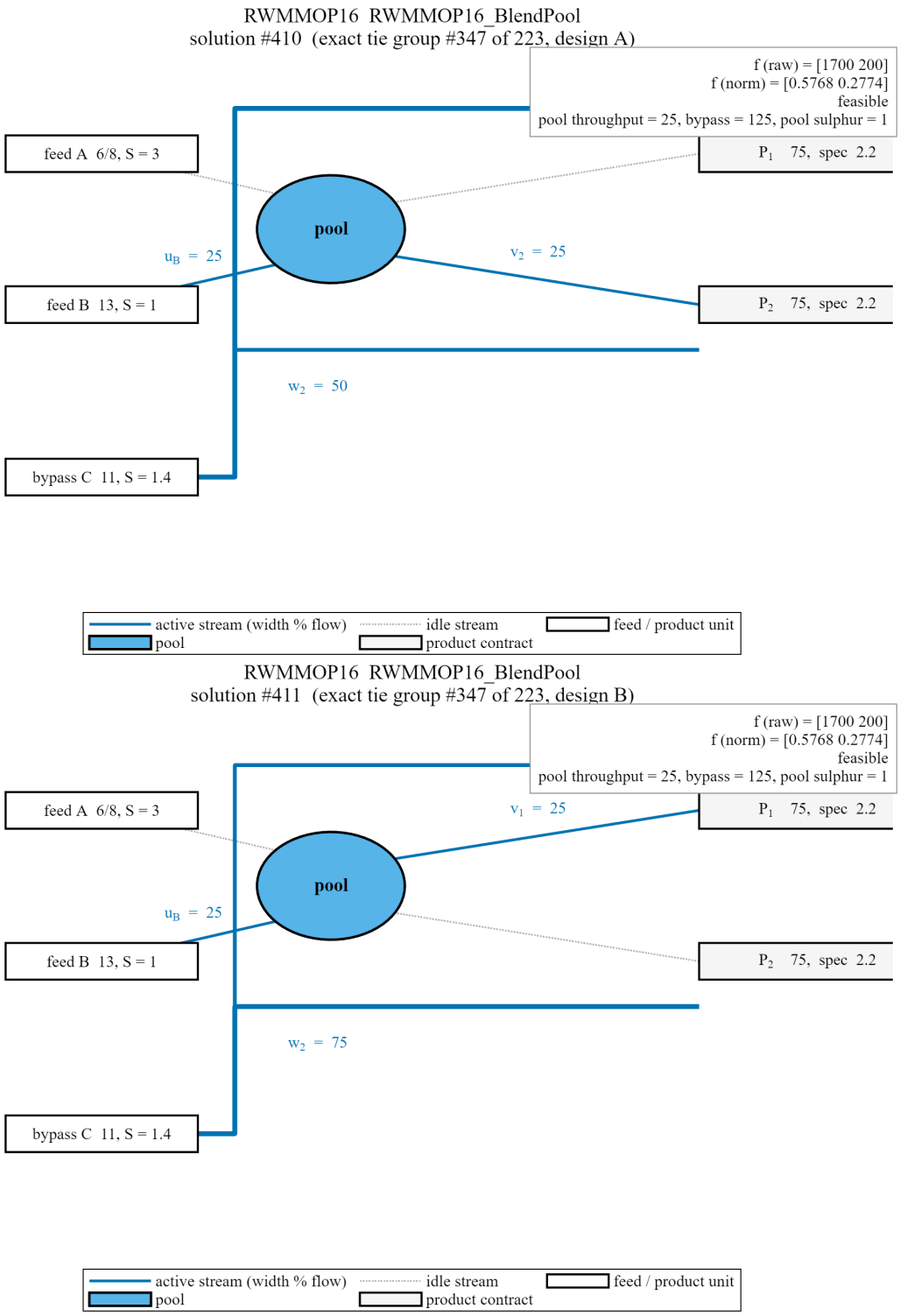
RWMMOP13 **DNR33MP** (GLPF): A (top) is a global optimum (front row 5); B (bottom) is a certified local optimum (certificate row 32) — the most decision-distant certificate row within twice the nearest-objective gap, objectives only $\Delta f = 0.137$ from A's: $(f_1, f_2, f_3) = (141.916,\ 0.0756302,\ 0)$ vs $(142.429,\ 0.0766583,\ 0)$. Yet all 2 decision coordinates ($\{1, 2\}$) differ (decision distance 0.198). Both period topologies change (the two topology indices flip); the local pair costs only 0.51 kW more loss with zero extra switching.



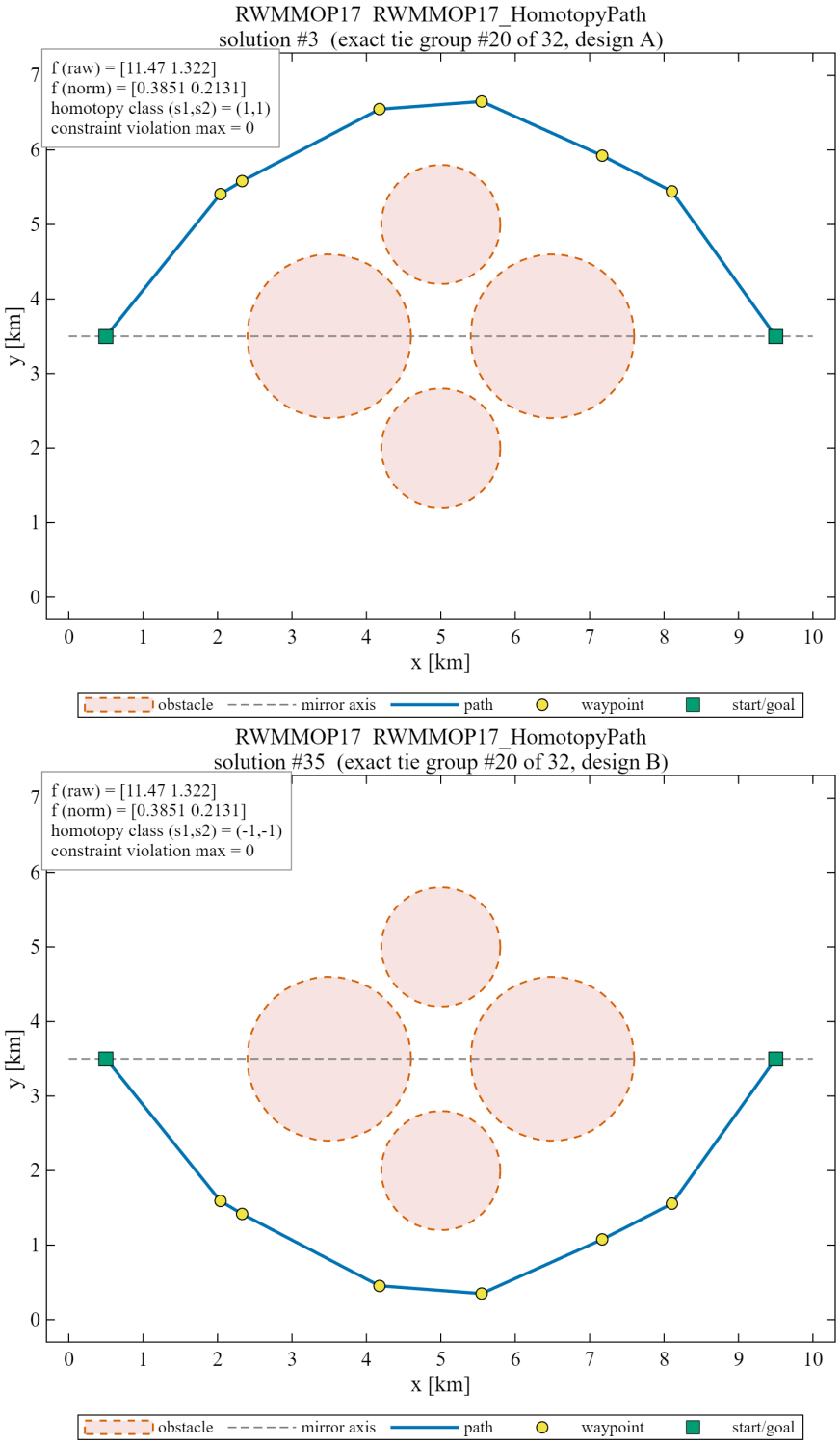
RWMMOP14 **Net12Sparse** (GLPF): A (top) is a global optimum (front row 16); B (bottom) is a certified local optimum (certificate row 2413) — the most decision-distant certificate row within twice the nearest-objective gap, objectives only $\Delta f = 0.0038$ from A's: $(f_1, f_2) = (27.9049,\ 201)$ vs $(27.9033,\ 202)$. Yet 4 of the 21 decision coordinates ($\{3, 7, 11, 13\}$) differ (decision distance 2). Four link states differ; the certified local design even shaves 0.0015 off the loss objective but pays one extra active link (202 vs 201) — a near miss off the global front.



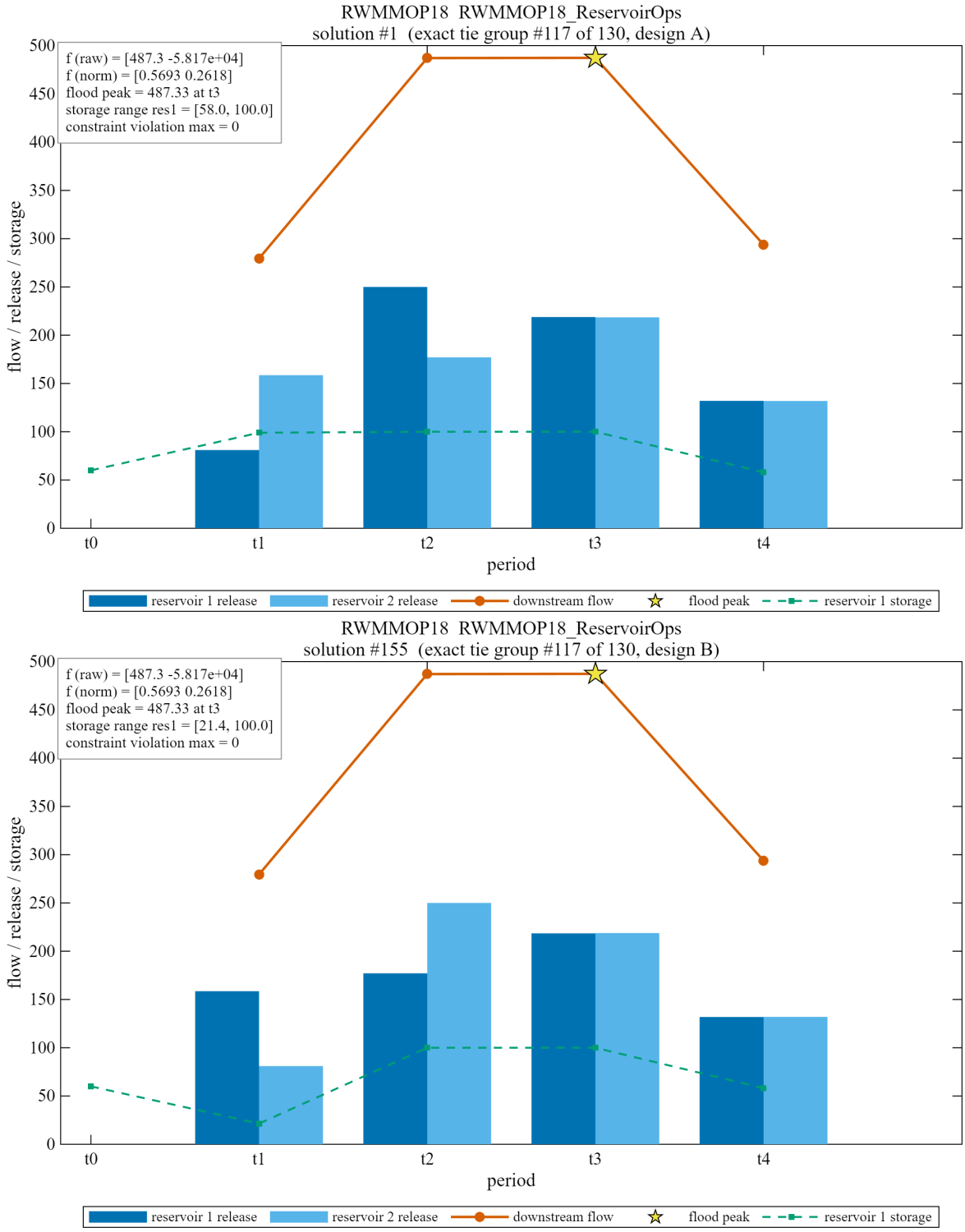
RWMMOP15 **SiteSelect** (MIX): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #396 (2 members in the published archive, global rows 726 and 1684). Objectives bit-identical, $(f_1, f_2, f_3) = (1.05055,\ 1.88882,\ 0.0254951)$; yet 1 of the 2 decision coordinates ($\{1\}$) differ (decision distance 0.682). B is the corridor-mirror image of A: the site's first coordinate flips while the second stays; coverage, mean distance and corridor cost are bit-identical.



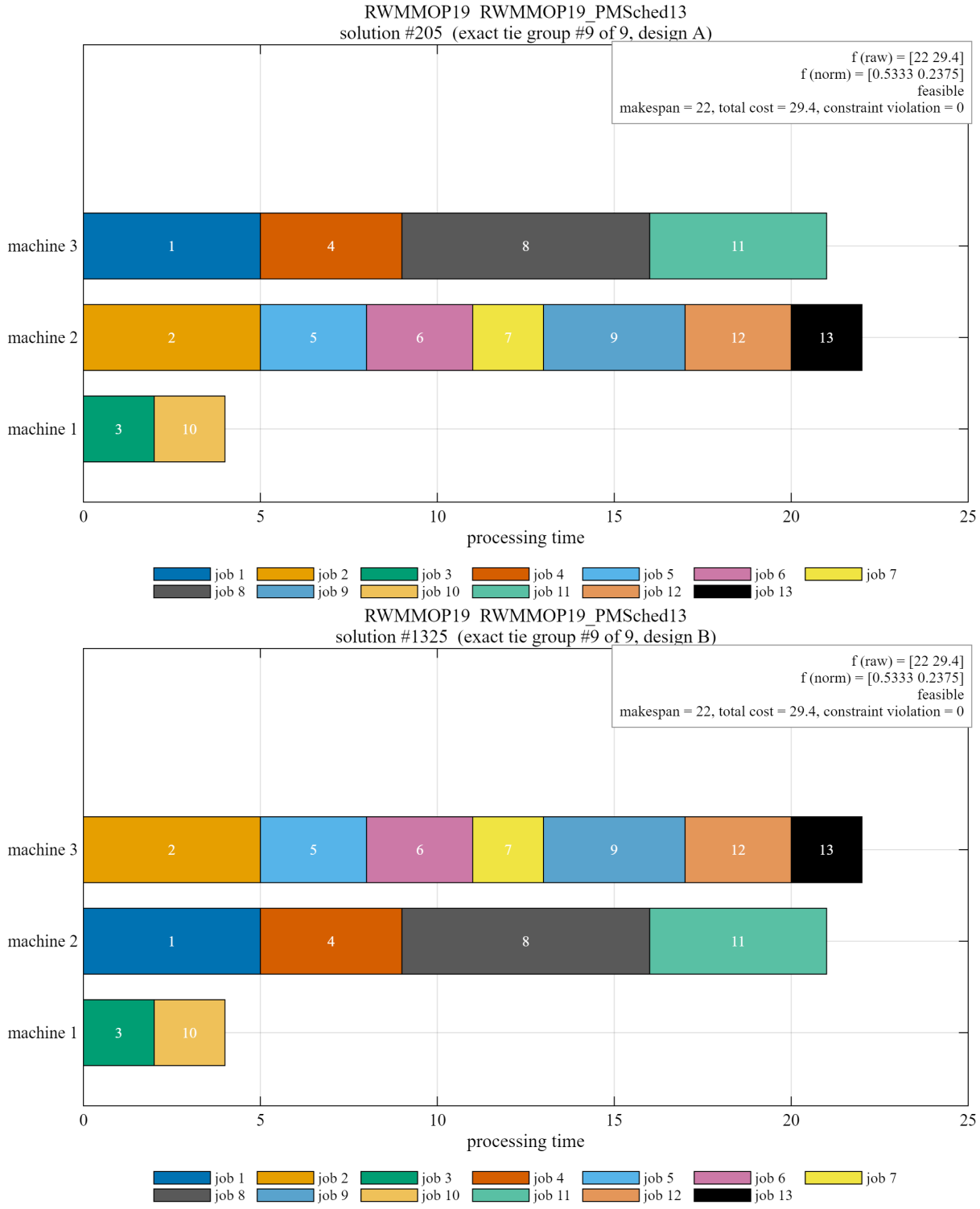
RWMMOP16 **BlendPool** (MIX): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #347 (2 members in the published archive, global rows 410 and 411). Objectives bit-identical, $(f_1, f_2) = (1700, 200)$; yet 1 of the 3 decision coordinates ($\{3\}$) differ (decision distance 0.333). Only the pool-to-P1 valve setting changes — v1 shifts by one 25-unit detent, a neutral dimension on this stretch; feed cost and product purity are bit-identical.



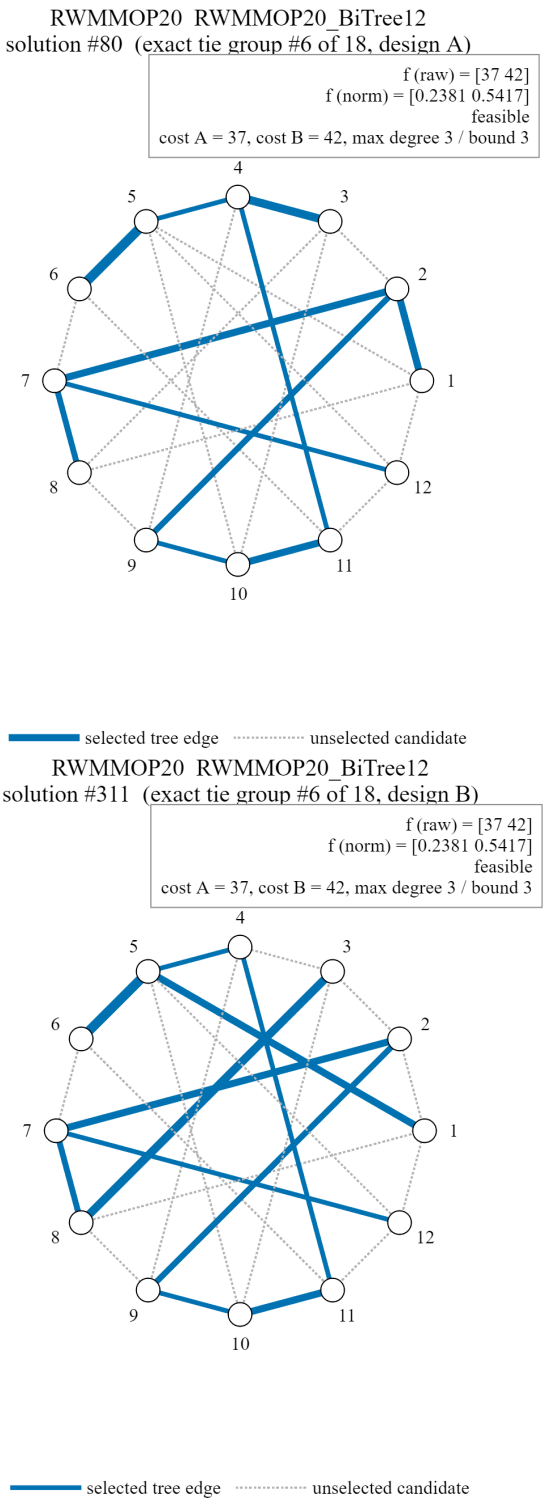
RWMMOP17 **HomotopyPath** (MIX): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #20 (2 members in the published archive, global rows 3 and 35). Objectives bit-identical, $(f_1, f_2) = (11.4706,\ 1.32209)$; yet 6 of the 12 decision coordinates ($\{2, 4, 6, 8, 10, 12\}$) differ (decision distance 1.94). All six lateral waypoint coordinates change: the two paths round the obstacles on the opposite side (two homotopy classes) at identical length and clearance.



RWMMOP18 **ReservoirOps** (MIX): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #117 (2 members in the published archive, global rows 1 and 155). Objectives bit-identical, $(f_1, f_2) = (487.332, -58172.1)$; yet all 8 decision coordinates ($\{1, 2, 3, 4, 5, 6, 7, 8\}$) differ (decision distance 1.38). Every period's release changes (all eight coordinates), yet the two operation rules return identical objective values.



RWMMOP19 **PMSched13** (MIX): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #9 (133 members in the published archive, global rows 205 and 1325). Objectives bit-identical, $(f_1, f_2) = (22, 29.4)$; yet 11 of the 13 decision coordinates ($\{1, 2, 4, 5, 6, 7, 8, 9, 11, 12, 13\}$) differ (decision distance 1.87). Eleven of the thirteen job-to-machine assignments change at identical makespan and cost — two of the 133 archived equivalent schedules on this front point.



RWMMOP20 **BiTree12** (MIX): designs A (top) and B (bottom) are the most render-distinct pair of exact tie group #6 (6 members in the published archive, global rows 80 and 311). Objectives bit-identical, $(f_1, f_2) = (37, 42)$; yet 4 of the 24 decision coordinates ($\{1, 3, 13, 15\}$) differ (decision distance 2). Four edge choices (1, 3, 13 and 15) reroute the spanning tree at identical values of both objectives.

S3. Reference Archives and Variable Layouts

Every problem stores its reference in two frozen MATLAB archives under the problem directory: a *global* archive (the Pareto front with one or more decision preimages per front point) and, where a trap axis exists, a *local* archive (the quality-gated local-optimum certificate). Both carry the frozen objective normalisation constants used by every metric, and both are SHA-256-bound in the data manifest (129 entries). Published archives are fixed points of the mode filter (main paper, Section III-C): within an objective-equivalence class only rows pairwise farther than $\delta_{\mathrm{far}} = 0.2$ (archive-range normalised) survive, one representative per cluster, local certificates keeping the global row first. The filter only deletes rows, so front coverage, IGD/HV denominators and the normalisation constants are unchanged by publication; pre-filter originals are retained separately as completeness evidence. Variable semantics (decision-vector layout, units, repair semantics for discrete encodings) are documented per problem in the catalogue; encodings are real (9 problems), binary (5), integer (5) and permutation/integer combined (1), as listed in Table S1.

S4. Licence and Provenance Audit

All 20 official problems carry licence status *verified* in the audited licence file, with the evidence trail per problem (public IEEE bus cases, public-domain classical models reimplemented from their defining papers, or self-constructed instances) retained in the release. No problem depends on code that may not be redistributed. The six ported *algorithms* (not problems) have the following status at the time of writing: CMDCoEA is the workspace owner's own algorithm and ships freely; the five third-party ports (CMMOCD, CMMOCEA, CMMODE, DCMMEA, DRCEA) are research-use with redistribution pending rights-holder confirmation, every ported file keeps its original authorship header, and the adaptation list plus SHA-256 of each of the 35 ported files is machine-checked by the regression suite. The data-availability statement of the main paper reflects the ruling at submission time; the benchmark's problem set and all results are independent of this ruling.

Provenance: every semantic file (problem classes, evaluators, frozen snapshots, reference archives) is hash-bound in the data manifest, and all 3714 result files of the pilots and the full eval_v2 round (3000 runs) are individually SHA-256-provenanced against the manifest version under which they were produced.

S5. Independent Oracles and Cross-Checks

Each evaluator family is cross-checked against an independently written second implementation before its archive is trusted:

- *Power flow (DNR family).* The distribution-network evaluator is differenced against a separately implemented Newton–Raphson solver (agreement $\leq 2.5 \times 10^{-12}$ pu), and the RWMMOP11 archive is the complete enumeration of all 407,924 feasible spanning trees, with the tree count itself cross-checked by a Kirchhoff matrix-tree computation.
- *Facility location (RWMMOP15).* A Node.js re-implementation of the distance geometry agrees with the MATLAB evaluator to 8.9×10^{-16}; the certified-dense archive carries an ε-density certificate and a Hausdorff bound of 0.0335 km per axis.
- *Weapon-target assignment (RWMMOP9).* An independently coded loop-based evaluator agrees to 8.9×10^{-17} with exact integer counts equal.
- *Truss (RWMMOP10).* An independently assembled stiffness-method solver agrees to 3.7×10^{-16} in objectives and 2.5×10^{-12} in stresses; the mirror symmetry that generates the equivalent sets is verified numerically to 3.4×10^{-16}.
- *Scheduling and combinatorial families.* Exact-enumeration problems (RWMMOP19: 3^{13}; RWMMOP8: 2^{24}; RWMMOP20: all spanning trees; RWMMOP14: 2^{21}) validate evaluators by construction, since the archive is the enumeration; satellite evaluators are audited on instance data (orbit and visibility windows) with the neutral-dimension finding recorded (demands 19–21 of RWMMOP1 have zero execution frequency).

The oracle identities, tolerances and per-problem evidence paths are part of the machine-generated catalogue and are asserted, not narrated, by the regression suite. The Node.js re-implementation of the RWMMOP15 distance geometry (and the other second implementations above) serves differential verification at archive-build time only; none of them is a runtime dependency of the benchmark or of the evaluation pipeline.

S6. Full Statistical Tables (eval_v2)

This section tabulates the full eval_v2 results. Tables S2–S5 give the per-problem medians and IQRs for all four metrics (generated by the evaluation pipeline; no cell is hand-typed); Table S6 adds the aggregate Friedman ranks. The tabulation conventions follow.

Conventions. A run whose final population contains no feasible solution records NaN for IGD/HV (platform contract); 118 of 3000 runs are such, always aligned with a zero feasible rate. Statistics are tabulated under three explicit lenses: *as-recorded* (NaNs kept), *feasibility-gated* (problem–algorithm pairs whose ten runs are all infeasible are excluded from IGD/IGDX/HV comparison; ten such pairs exist), and *penalised* (such pairs receive the worst rank). Rankings in the main paper use the penalised lens. Friedman mean ranks and the Nemenyi CD ($\alpha = 0.05$, $k = 15$, $N = 20$ problems) use the complete matrix; the Friedman null hypothesis is that all fifteen algorithms share one rank distribution over the twenty problems, and the Nemenyi CD controls the family-wise error of all pairwise rank comparisons at $\alpha = 0.05$. Pairwise Wilcoxon signed-rank tests (two-sided, one per problem and metric, null: zero median difference against the per-problem best) are reported with rank-biserial effect sizes $r_{rb} = 2W^+/(n(n+1)) - 1$ in the released CSV.

Table S6 makes the main paper's two-axis reading checkable in one place: the IGDX column is led by the five ported constrained-MMO algorithms (DCMMEA 5.33, CMMOCD 5.80, DRCEA 5.70, CMMOCEA 6.48, plus CMMODE at 8.85), while the feasibility column is led by CoMMEA (3.93), MOEA/D (4.25) and RVEA (5.03). The full post-hoc artefacts (feasibility-gate table, fair-wins table, first-hit table, budget slices, per-problem axis wins, per-seed medians and the Wilcoxon / rank-biserial columns) ship as CSV files with the release and are regenerated by `eval_v2_posthoc.m` in seconds.

S7. Trajectory Grids and Gallery Indexes

The full-resolution convergence grid of the whole panel appears as main-paper Fig. 8; this section keeps the budget-slice heat maps and the gallery indexes.

In addition to the static figures, two interactive HTML galleries ship with the release: the *domain gallery* (one rendered engineering solution per figure, 152 figures, deterministic showcase rules: largest tie group in full, front extremes, knee, local witness) and the *multimodal-evidence gallery* (per problem: front with tie groups coloured, decision-space tie structure, separation scatter). Both are regenerated by scripts from the frozen archives. A static per-problem equivalent-design showcase (two optima per problem, rendered by the same domain renderers) is Section S2 of this supplement.

S8. MMI Robustness Studies

Table S7 is the standard-problem calibration of MMI: analytic-PS and grid cases with the MMI value and the structure the literature assigns to each case. Three properties matter for the main paper's claims. *Semantic agreement*: unimodal cases score exactly 0, two-branch cases ≈ 1, and SYM-PART's nine segments score 12.05 (recorded far-field reading 8.00, i.e. the literature count minus one, confirming that the far gate removes fold artefacts). *Density robustness*: MMF1 holds 0.95 from 200 to 12,800 reference rows; IDMP grid references converge from 0.967 to 0.996 as the grid refines from 301 to 2401 points. *Adaptive-tolerance veto*: allowing the equivalence tolerance to adapt ($\alpha > 0$) created false positives on sparse archives (RWMMOP11: $0 \to 0.33$; RWMMOP14: $0 \to 0.35$), which is why the fixed $\varepsilon = 10^{-6}$ is the published parameter and the sweep is retained as negative evidence. The δ sensitivity sweep over the full problem set and the near/far verdict-boundary analysis are part of the released `mmi_study` artefacts.

S9. Reproduction Instructions

Environment. MATLAB (the study runs used R2025b on Windows and R2026a on Linux for the main evaluation) with the Statistics and Machine Learning Toolbox and the Parallel Computing Toolbox; PlatEMO 4.x on the path. The regression suite is platform-independent; batch experiments (more than twenty runs) run locally on a twenty-worker parallel pool with per-run on-disk caching (checkpointed resume).

One-command validation.

```
addpath(genpath('<PlatEMO>'));
addpath('<repo>/tests');
run_regression_suite
```

rebuilds and re-checks everything bounded: certificate consistency, archive fixed-point property, manifest hashes, gallery and report artefacts, generated paper tables, and smoke evaluations. The suite currently has 123 checks and must be all green; the item count is read from the generated summary CSV, never hard-typed. Full experiment reproduction runs the same evaluation entry point the study used (`eval_v2_experiment`) with the released manifest: 20 problems, 15 algorithms, 10 seeds, per-generation trajectories, a 20-worker pool, per-run on-disk caching (resume-safe), and per-run determinism via `rng(seed,'twister')` — identical results whether serial or parallel.

What is checked, not trusted. Every generated table and figure in the paper and this supplement has a validator: the paper-table generator re-derives each cell from the metadata JSONs inside the regression suite; the paper-index validator asserts that every evidence path cited by the paper's claim table exists on disk; the report validator compares every headline number in the HTML report against the CSVs. A number that cannot be regenerated is treated as a bug, not a footnote.

Table S2: IGD per problem (median over repetitions, IQR in parentheses; best in **bold**). Budget is problem-dependent (maxFE = min(max(5000D,20000),50000)); approximate references carry their caveat.

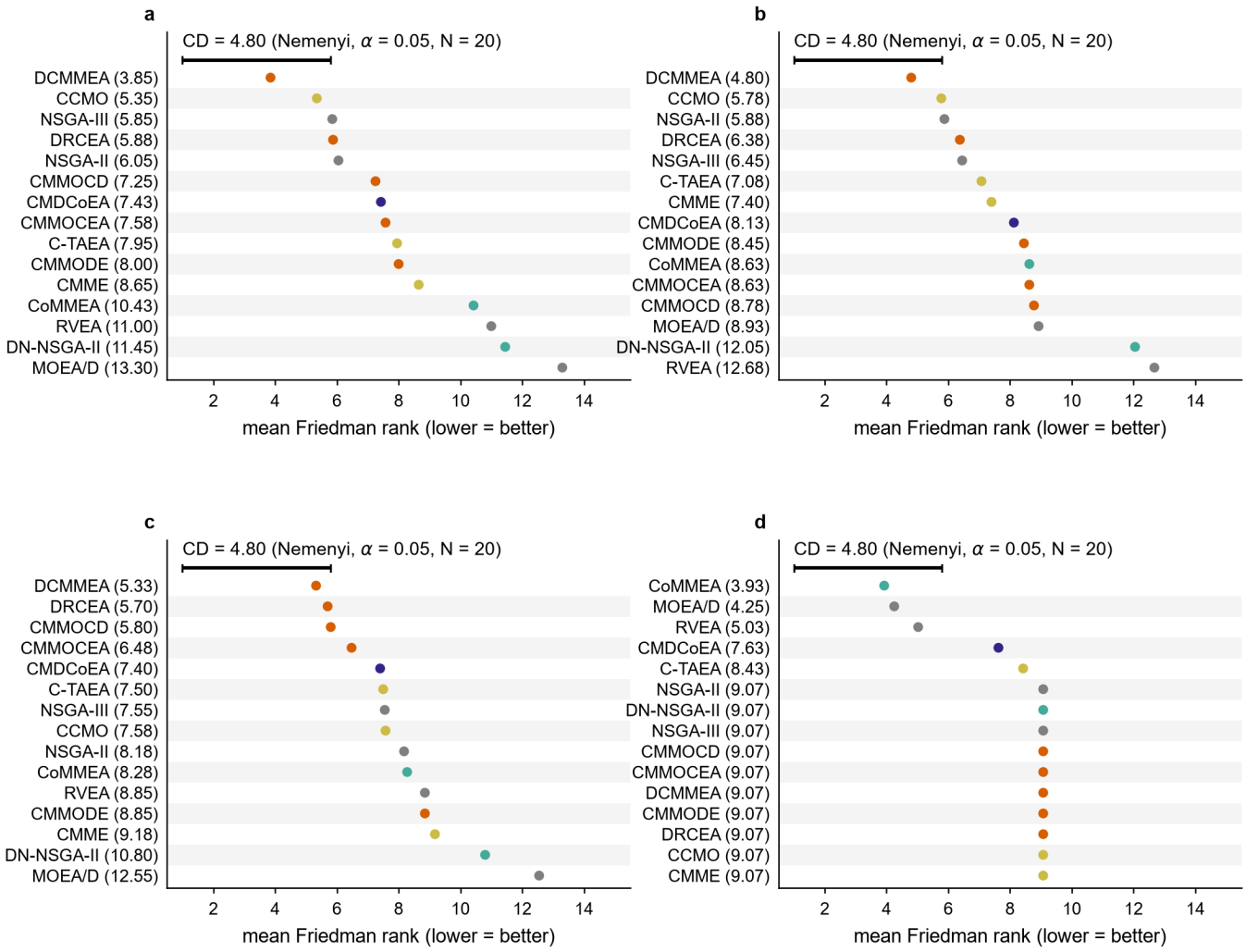
Problem	NSGAII	NSGAIII	MOEAD	RVEA	CoMMEA	DNNSGAII	CMDCoEA	CMMOCD	CMMOCEA	CMMODE	DCMMEA	DRCEA	CCMO	CMME	CTAEA
DNR69	**0.000e+00 (5.4e-04)**	1.775e-03 (0.0e+00)	3.595e-01 (7.1e-01)	5.716e-01 (0.0e+00)	2.204e-01 (0.0e+00)	5.896e-01 (0.0e+00)	1.254e+00 (0.0e+00)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	7.528e-02 (1.4e-01)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	3.470e-03 (0.0e+00)	3.470e-03 (0.0e+00)
DNR33MS	**0.000e+00 (0.0e+00)**	0.000e+00 (0.0e+00)	2.126e-02 (1.5e-01)	2.608e-01 (3.1e-02)	0.000e+00 (0.0e+00)	5.351e-01 (4.7e-01)	2.389e-02 (2.1e-02)	4.953e-03 (1.4e-02)	0.000e+00 (5.1e-03)	1.398e-01 (4.9e-02)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	5.130e-03 (0.0e+00)	5.130e-03 (0.0e+00)
DNR33MP	**0.000e+00 (8.0e-02)**	0.000e+00 (8.0e-02)	2.553e-01 (2.9e-02)	3.356e-02 (1.7e-02)	0.000e+00 (0.0e+00)	7.966e-02 (8.0e-02)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	0.000e+00 (8.0e-02)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	0.000e+00 (8.0e-02)	3.983e-02 (8.0e-02)	0.000e+00 (0.0e+00)
SatelliteTTC25	1.113e-01 (5.3e-02)	1.109e-01 (5.2e-02)	5.316e-01 (1.1e-01)	1.342e-01 (9.4e-02)	**5.924e-02 (5.4e-03)**	4.496e-01 (1.7e-01)	6.530e-02 (7.5e-03)	1.767e-01 (7.6e-02)	3.289e-01 (6.8e-02)	1.115e-01 (5.7e-02)	8.931e-02 (2.9e-02)	8.492e-02 (3.2e-02)	1.035e-01 (8.4e-03)	2.361e-01 (1.0e-01)	1.285e-01 (4.3e-02)
Jobshop5	**5.475e-03 (5.8e-04)**	6.762e-03 (5.9e-04)	2.315e-02 (1.0e-02)	1.569e-02 (2.4e-03)	1.978e-02 (1.0e-03)	8.102e-02 (4.2e-02)	6.025e-03 (6.8e-04)	1.069e-02 (6.1e-04)	1.447e-02 (3.1e-03)	1.285e-02 (4.0e-03)	7.587e-03 (1.1e-03)	9.108e-03 (1.6e-03)	7.084e-03 (1.5e-03)	6.626e-03 (8.3e-04)	8.997e-03 (7.5e-04)
SatelliteTTC50	3.143e-01 (3.9e-02)	2.497e-01 (4.9e-02)	5.814e-01 (1.2e-01)	1.839e-01 (4.8e-02)	**9.500e-02 (2.3e-02)**	5.301e-01 (1.7e-01)	1.848e-01 (5.7e-02)	1.700e-01 (5.5e-02)	2.013e-01 (5.8e-02)	1.638e-01 (3.3e-02)	2.092e-01 (9.3e-02)	3.002e-01 (1.6e-01)	2.098e-01 (4.6e-02)	3.764e-01 (9.9e-02)	2.807e-01 (1.5e-01)
SiteSelect	4.677e-02 (6.3e-03)	4.863e-02 (5.3e-03)	7.237e-02 (4.6e-03)	8.096e-02 (5.9e-03)	6.544e-02 (4.0e-03)	5.235e-02 (3.9e-03)	4.317e-02 (4.5e-03)	4.022e-02 (9.7e-04)	4.011e-02 (1.8e-03)	4.404e-02 (2.5e-03)	3.686e-02 (7.3e-04)	3.667e-02 (1.2e-03)	**3.459e-02 (9.0e-04)**	4.704e-02 (6.2e-03)	4.669e-02 (4.5e-03)
BlendPool	7.003e-03 (6.4e-04)	6.140e-03 (2.1e-03)	7.738e-02 (7.5e-02)	1.138e-02 (2.2e-03)	3.629e-02 (6.9e-03)	6.872e-03 (3.3e-04)	**4.140e-03 (2.0e-03)**	7.304e-03 (2.1e-03)	7.702e-03 (4.3e-04)	7.319e-03 (7.6e-04)	5.701e-03 (2.1e-03)	4.968e-03 (1.9e-03)	5.649e-03 (4.7e-05)	6.589e-03 (2.1e-03)	5.722e-03 (2.0e-03)
ArrayNull	3.571e-02 (1.5e-02)	4.239e-02 (6.4e-02)	1.677e-01 (2.3e-01)	1.053e-01 (4.2e-02)	8.853e-02 (1.7e-02)	5.442e-02 (6.1e-02)	2.783e-02 (5.7e-02)	4.330e-02 (1.8e-02)	3.271e-02 (2.7e-02)	**1.556e-02 (4.1e-03)**	3.110e-02 (2.0e-02)	3.549e-02 (2.0e-02)	5.473e-02 (1.4e-02)	3.340e-02 (2.1e-02)	4.661e-02 (1.7e-02)
WindFarm	2.608e-01 (5.2e-01)	1.860e-01 (2.2e-01)	n/a	5.310e-01 (3.4e-01)	8.481e-01 (1.7e+00)	2.315e-01 (2.0e-01)	9.384e-01 (1.0e+00)	6.560e-01 (2.8e-01)	3.167e-01 (3.3e-01)	2.740e-01 (2.9e-01)	2.061e-01 (2.5e-01)	5.747e-01 (6.0e-01)	2.293e-01 (1.0e-01)	2.536e-01 (1.7e-01)	**1.572e-01 (1.0e-01)**
HomotopyPath	9.857e-02 (4.0e-03)	9.997e-02 (6.4e-03)	1.122e-01 (8.4e-03)	**7.768e-02 (1.2e-02)**	8.377e-02 (4.0e-03)	8.636e-02 (1.1e-02)	1.019e-01 (9.3e-03)	9.175e-02 (7.7e-03)	9.680e-02 (2.7e-03)	1.042e-01 (9.9e-03)	1.003e-01 (6.6e-03)	9.305e-02 (5.3e-03)	1.016e-01 (5.1e-03)	1.033e-01 (9.5e-03)	1.078e-01 (3.5e-03)
NestingLayout	**2.551e-02 (2.1e-02)**	3.166e-02 (7.7e-03)	n/a	1.085e-01 (1.9e-02)	n/a	5.374e-02 (5.2e-02)	2.319e-01 (2.6e-01)	6.018e-02 (2.1e-02)	5.563e-02 (2.0e-02)	4.290e-02 (3.4e-02)	3.123e-02 (1.6e-02)	4.919e-02 (1.7e-02)	4.538e-02 (1.0e-02)	5.872e-02 (1.8e-02)	3.218e-01 (3.7e-02)
ReservoirOps	1.943e-02 (1.8e-03)	1.922e-02 (1.5e-03)	n/a	7.657e-02 (1.5e-02)	n/a	2.252e-02 (1.4e-03)	1.160e-01 (1.0e-01)	2.143e-02 (8.8e-04)	2.182e-02 (3.7e-04)	2.449e-02 (6.7e-03)	**1.852e-02 (1.8e-03)**	1.918e-02 (1.4e-03)	1.938e-02 (5.4e-04)	2.022e-02 (3.5e-03)	1.761e-01 (8.7e-02)
EcoDispatch15MF	8.234e-03 (2.1e-03)	9.209e-03 (6.5e-03)	6.505e-03 (1.3e-03)	2.756e-02 (7.2e-03)	3.020e-02 (1.8e-03)	3.344e-02 (1.8e-02)	1.152e-02 (5.0e-04)	5.522e-02 (5.2e-02)	8.894e-02 (2.6e-02)	1.020e-02 (9.6e-04)	8.829e-03 (7.8e-04)	1.614e-02 (7.1e-03)	**5.726e-03 (7.5e-04)**	1.517e-02 (6.4e-03)	1.429e-02 (8.5e-03)
PMSched13	1.550e-03 (2.1e-03)	6.027e-02 (7.2e-02)	4.900e-03 (1.1e-02)	3.211e-04 (1.6e-04)	7.137e-03 (1.1e-02)	4.391e-04 (8.7e-04)	3.112e-04 (3.4e-04)	3.877e-04 (1.1e-04)	8.441e-04 (6.7e-04)	6.544e-05 (4.3e-05)	**4.025e-05 (3.0e-05)**	2.604e-03 (2.1e-03)	1.984e-03 (2.1e-03)	3.802e-04 (8.0e-05)	
Knapsack24	5.193e-03 (8.2e-04)	**3.628e-03 (1.4e-04)**	8.651e-03 (1.5e-04)	1.140e-02 (8.5e-04)	1.896e-02 (1.6e-03)	1.010e-01 (6.2e-02)	4.131e-03 (3.7e-04)	1.039e-02 (5.2e-03)	1.478e-02 (4.3e-03)	7.307e-03 (1.1e-03)	5.188e-03 (3.5e-04)	5.231e-03 (2.7e-04)	4.098e-03 (2.5e-04)	4.113e-03 (3.4e-04)	4.371e-03 (2.5e-04)
BiTree12	3.290e-03 (3.6e-03)	**1.636e-03 (3.5e-03)**	n/a	6.315e-02 (4.3e-02)	n/a	1.560e-01 (2.6e-01)	n/a	1.829e-03 (2.4e-03)	4.288e-03 (2.2e-03)	1.568e-02 (2.6e-02)	3.790e-03 (3.6e-03)	6.446e-03 (3.6e-03)	1.133e-02 (1.3e-02)	7.463e-03 (6.0e-03)	5.352e-02 (1.1e-02)
Net12Sparse	4.603e-03 (2.7e-03)	3.259e-03 (3.1e-03)	n/a	8.361e-02 (1.4e-02)	n/a	1.385e-01 (7.6e-02)	7.984e-01 (6.3e-01)	3.037e-03 (2.6e-03)	4.242e-03 (8.5e-03)	2.274e-02 (1.2e-02)	3.037e-03 (1.6e-03)	3.820e-03 (2.7e-03)	4.603e-03 (1.1e-03)	3.881e-03 (1.6e-03)	**2.900e-03 (2.0e-03)**
WTA66	1.781e-02 (2.5e-02)	1.724e-02 (7.1e-03)	7.502e-02 (2.7e-02)	3.661e-02 (1.9e-02)	4.556e-02 (3.2e-02)	3.662e-01 (1.8e-01)	**3.397e-03 (3.0e-03)**	1.252e-02 (1.2e-02)	9.519e-03 (7.0e-03)	2.121e-02 (9.1e-03)	3.488e-03 (2.5e-03)	9.889e-03 (1.0e-02)	5.099e-03 (2.7e-03)	2.075e-02 (1.4e-02)	3.058e-02 (2.9e-02)
Truss25	1.559e-02 (2.1e-02)	7.158e-02 (3.5e-02)	3.984e-02 (6.2e-03)	6.257e-02 (1.2e-02)	2.971e-01 (4.9e-02)	3.661e-02 (9.5e-03)	1.453e-02 (1.3e-03)	3.968e-02 (1.9e-02)	3.630e-02 (1.6e-02)	1.473e-02 (6.3e-03)	**1.008e-02 (3.4e-03)**	9.460e-02 (5.0e-02)	1.326e-02 (4.2e-03)	1.478e-01 (1.1e-01)	2.611e-02 (1.3e-02)

Table S3: IGDX per problem (median over repetitions, IQR in parentheses; best in **bold**). Budget is problem-dependent (maxFE = min(max(5000D,20000),50000)); approximate references carry their caveat.

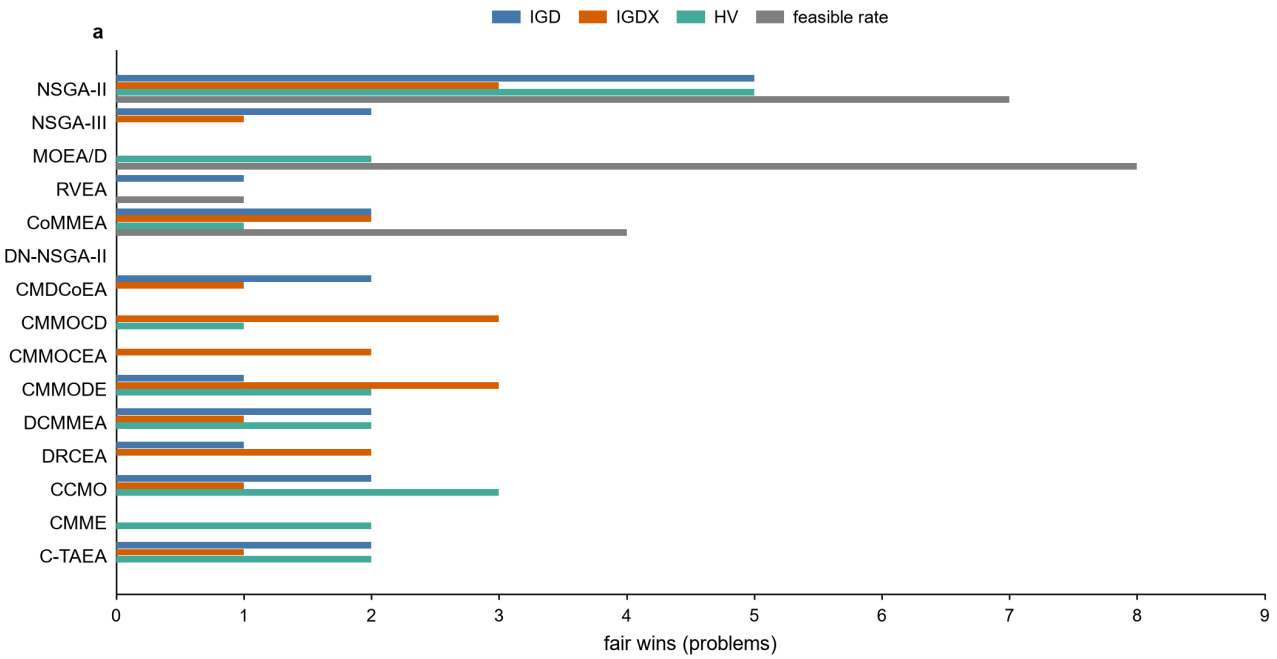
Problem	NSGAII	NSGAIII	MOEAD	RVEA	CoMMEA	DNNSGAII	CMDCoEA	CMMOCD	CMMOCEA	CMMODE	DCMMEA	DRCEA	CCMO	CMME	CTAEA
DNR69	**0.000e+00 (2.4e-01)**	4.714e-01 (0.0e+00)	1.414e+00 (1.4e-01)	1.179e+00 (0.0e+00)	7.071e-01 (0.0e+00)	1.794e+00 (0.0e+00)	1.569e+00 (0.0e+00)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	3.170e+00 (1.3e-01)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	7.071e-01 (0.0e+00)	7.071e-01 (0.0e+00)
DNR33MS	**0.000e+00 (0.0e+00)**	0.000e+00 (0.0e+00)	2.357e-01 (8.0e-01)	1.618e+00 (1.3e-01)	0.000e+00 (0.0e+00)	2.083e+00 (1.2e-01)	1.179e-01 (1.2e-01)	1.179e-01 (1.2e-01)	0.000e+00 (1.2e-01)	2.662e+00 (1.4e-01)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	1.179e-01 (0.0e+00)	1.179e-01 (0.0e+00)
DNR33MP	**0.000e+00 (3.3e+00)**	0.000e+00 (3.3e+00)	2.837e+00 (5.3e+00)	1.429e-01 (5.9e-02)	0.000e+00 (0.0e+00)	3.286e+00 (3.3e+00)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	2.608e-01 (3.1e+00)	0.000e+00 (0.0e+00)	0.000e+00 (0.0e+00)	0.000e+00 (3.3e+00)	1.643e+00 (3.3e+00)	0.000e+00 (0.0e+00)
SatelliteTTC25	3.756e+01 (7.3e-01)	3.757e+01 (9.3e-01)	3.872e+01 (5.1e-01)	3.765e+01 (1.2e+00)	3.722e+01 (2.9e-01)	3.949e+01 (1.7e+00)	3.654e+01 (8.9e-01)	4.142e+01 (1.6e+00)	4.144e+01 (1.1e+00)	**3.554e+01 (1.2e+00)**	3.764e+01 (8.2e-01)	3.877e+01 (8.6e-01)	3.819e+01 (7.0e-01)	3.827e+01 (1.9e+00)	3.772e+01 (1.2e+00)
Jobshop5	6.985e+00 (2.8e-01)	6.938e+00 (2.6e-01)	7.152e+00 (1.4e-01)	6.886e+00 (3.8e-01)	6.646e+00 (1.0e-01)	8.018e+00 (2.6e-01)	**6.590e+00 (1.4e-01)**	7.080e+00 (6.8e-02)	7.136e+00 (1.2e-01)	7.842e+00 (4.8e-01)	6.990e+00 (1.1e-01)	7.087e+00 (4.7e-02)	7.191e+00 (1.0e-01)	6.855e+00 (3.3e-01)	7.157e+00 (3.3e-01)
SatelliteTTC50	5.728e+01 (4.4e-01)	5.711e+01 (6.5e-01)	5.818e+01 (4.9e-01)	5.628e+01 (1.3e+00)	**5.537e+01 (8.0e-01)**	5.890e+01 (1.4e+00)	5.690e+01 (6.9e-01)	5.699e+01 (7.6e-01)	5.719e+01 (1.0e+00)	5.757e+01 (1.1e+00)	5.705e+01 (1.4e+00)	5.785e+01 (1.9e+00)	5.724e+01 (1.2e+00)	5.816e+01 (1.2e+00)	5.703e+01 (1.6e+00)
SiteSelect	1.409e-01 (4.6e-02)	1.543e+00 (2.6e+00)	8.614e-01 (8.2e-01)	3.053e-01 (1.5e+00)	1.405e-01 (1.0e-02)	1.269e-01 (1.1e-02)	1.030e-01 (5.1e-03)	8.791e-02 (2.2e-03)	**8.783e-02 (3.9e-03)**	1.382e-01 (6.1e-02)	9.085e-02 (2.9e-03)	8.975e-02 (2.3e-03)	1.120e-01 (4.0e-02)	1.630e+00 (2.7e+00)	2.063e-01 (1.3e-01)
BlendPool	5.905e+00 (1.1e+00)	5.111e+00 (1.8e+00)	1.727e+01 (6.0e+00)	5.827e+00 (7.4e-01)	9.058e+00 (1.3e+00)	3.604e+00 (1.2e+00)	3.237e+00 (1.5e+00)	2.064e+00 (7.7e-01)	2.030e+00 (7.6e-02)	6.540e+00 (8.1e-01)	2.019e+00 (7.2e-01)	**1.640e+00 (8.1e-01)**	4.004e+00 (2.2e+00)	5.701e+00 (2.9e+00)	5.210e+00 (1.9e+00)
ArrayNull	1.888e-01 (1.5e-01)	1.490e-01 (4.3e-02)	2.220e-01 (1.5e-01)	1.867e-01 (8.1e-02)	9.212e-02 (1.7e-02)	1.566e-01 (7.4e-02)	1.785e-01 (7.4e-02)	1.737e-01 (1.1e-01)	2.114e-01 (5.8e-02)	**8.088e-02 (3.6e-02)**	1.758e-01 (1.1e-01)	1.534e-01 (6.6e-02)	1.521e-01 (7.3e-02)	1.327e-01 (9.8e-02)	1.384e-01 (6.7e-02)
WindFarm	4.666e+00 (2.6e+00)	3.478e+00 (4.1e-01)	4.295e+00 (1.2e+00)	3.098e+00 (8.1e-01)	2.978e+00 (2.5e-01)	3.686e+00 (5.8e-01)	3.541e+00 (1.2e+00)	**2.759e+00 (3.7e-01)**	3.060e+00 (4.2e-01)	4.048e+00 (1.0e+00)	4.215e+00 (1.6e+00)	3.251e+00 (5.6e-01)	3.362e+00 (8.6e-01)	3.278e+00 (5.0e-01)	3.709e+00 (2.2e-01)
HomotopyPath	6.884e+00 (1.8e+00)	7.465e+00 (1.3e+00)	7.198e+00 (1.4e+00)	5.996e+00 (3.1e+00)	**3.471e+00 (1.3e+00)**	7.116e+00 (2.0e+00)	6.116e+00 (4.2e-01)	6.149e+00 (3.5e+00)	6.058e+00 (7.0e-01)	5.183e+00 (2.5e+00)	5.970e+00 (4.0e-01)	5.869e+00 (3.1e-01)	6.403e+00 (5.8e-01)	7.298e+00 (1.3e+00)	7.140e+00 (8.9e-01)
NestingLayout	1.495e+01 (9.8e-01)	1.466e+01 (1.7e+00)	1.952e+01 (1.5e+00)	1.301e+01 (1.2e+00)	1.151e+01 (2.4e-01)	1.499e+01 (1.9e+00)	1.405e+01 (1.5e+00)	1.339e+01 (6.2e-01)	1.363e+01 (1.8e+00)	1.439e+01 (1.1e+00)	1.426e+01 (1.3e+00)	1.279e+01 (9.3e-01)	1.509e+01 (2.0e+00)	1.555e+01 (2.0e+00)	**1.029e+01 (4.8e-01)**
ReservoirOps	8.997e+01 (3.8e+01)	9.711e+01 (1.3e+01)	1.551e+02 (3.1e+01)	8.424e+01 (1.0e+01)	1.282e+02 (7.1e+00)	8.443e+01 (3.0e+01)	8.170e+01 (4.5e+01)	4.056e+01 (9.7e+00)	5.372e+01 (4.0e+01)	5.992e+01 (3.2e+01)	8.809e+01 (1.7e+01)	**3.112e+01 (4.6e+00)**	9.211e+01 (2.3e+01)	1.047e+02 (2.3e+01)	9.552e+01 (3.3e+01)
EcoDispatch15MF	4.944e+01 (4.2e+00)	4.807e+01 (1.9e+00)	4.804e+01 (4.5e+00)	7.107e+01 (4.0e+00)	9.656e+01 (7.8e+00)	6.434e+01 (2.4e+00)	6.687e+01 (3.2e+00)	7.116e+01 (7.9e+00)	7.320e+01 (4.5e+00)	9.070e+01 (2.3e+01)	5.866e+01 (2.4e+00)	5.787e+01 (2.3e+00)	**4.581e+01 (1.9e+00)**	4.790e+01 (3.6e+00)	4.651e+01 (2.5e+00)
PMSched13	1.932e+00 (8.9e-02)	2.117e+00 (6.3e-02)	2.439e+00 (7.7e-02)	1.984e+00 (2.6e-02)	1.851e+00 (1.9e-02)	1.925e+00 (5.5e-02)	1.931e+00 (3.5e-02)	1.519e+00 (5.5e-02)	**1.518e+00 (1.5e-02)**	1.836e+00 (8.3e-02)	1.584e+00 (1.4e-01)	1.556e+00 (1.8e-02)	2.286e+00 (8.8e-02)	2.007e+00 (6.3e-02)	2.157e+00 (8.8e-02)
Knapsack24	1.188e+00 (9.1e-03)	**1.092e+00 (4.1e-03)**	1.398e+00 (3.3e-02)	1.641e+00 (1.7e-02)	2.139e+00 (9.1e-02)	1.382e+00 (1.0e-01)	1.158e+00 (4.0e-02)	1.602e+00 (5.4e-02)	1.648e+00 (4.2e-02)	1.645e+00 (6.1e-02)	1.365e+00 (1.9e-02)	1.376e+00 (2.4e-02)	1.130e+00 (4.9e-02)	1.117e+00 (2.2e-02)	1.116e+00 (1.4e-02)
BiTree12	1.522e+00 (1.3e-01)	1.365e+00 (1.1e-01)	3.162e+00 (0.0e+00)	2.167e+00 (2.5e-01)	3.317e+00 (0.0e+00)	2.055e+00 (6.7e-01)	2.579e+00 (6.9e-02)	**9.936e-01 (1.5e-02)**	1.003e+00 (1.8e-02)	1.747e+00 (4.9e-02)	9.996e-01 (1.1e-02)	1.023e+00 (3.1e-02)	1.421e+00 (8.6e-02)	1.482e+00 (1.0e-01)	1.992e+00 (9.0e-02)
Net12Sparse	3.529e-01 (2.6e-01)	2.769e-01 (1.5e-01)	3.180e+00 (0.0e+00)	1.368e+00 (1.3e-01)	3.334e+00 (0.0e+00)	1.706e+00 (1.3e-01)	1.994e+00 (7.3e-01)	**2.353e-01 (1.2e-01)**	2.769e-01 (2.4e-01)	1.374e+00 (8.6e-02)	2.353e-01 (8.3e-02)	2.941e-01 (1.5e-01)	3.529e-01 (1.2e-01)	3.185e-01 (1.2e-01)	2.597e-01 (1.2e-01)
WTA66	3.891e+00 (5.1e-01)	3.830e+00 (6.1e-01)	4.059e+00 (4.6e-01)	3.725e+00 (5.6e-01)	4.211e+00 (3.1e-01)	5.040e+00 (5.8e-01)	3.573e+00 (3.1e-01)	3.384e+00 (3.5e-01)	3.341e+00 (4.7e-01)	3.768e+00 (4.5e-01)	**3.301e+00 (1.4e-01)**	3.384e+00 (3.2e-01)	3.801e+00 (2.8e-01)	3.795e+00 (3.0e-01)	3.592e+00 (2.2e-01)
Truss25	8.617e-04 (1.0e-04)	1.108e-03 (2.5e-04)	9.191e-04 (9.9e-05)	1.695e-03 (1.6e-04)	2.958e-03 (7.1e-04)	9.471e-04 (8.4e-05)	9.492e-04 (3.2e-05)	1.111e-03 (1.8e-04)	1.190e-03 (1.3e-04)	**8.281e-04 (2.6e-05)**	8.826e-04 (3.4e-05)	1.405e-03 (3.1e-04)	9.278e-04 (3.7e-05)	1.373e-03 (6.6e-04)	8.934e-04 (8.1e-05)

Table S4: HV per problem (median over repetitions, IQR in parentheses; best in **bold**). Budget is problem-dependent (maxFE = min(max(5000D,20000),50000)); approximate references carry their caveat.

Problem	NSGAII	NSGAIII	MOEAD	RVEA	CoMMEA	DNNSGAII	CMDCoEA	CMMOCD	CMMOCEA	CMMODE	DCMMEA	DRCEA	CCMO	CMME	CTAEA
DNR69	**1.745e-01 (7.0e-06)**	1.727e-01 (0.0e+00)	9.681e-02 (2.8e-02)	9.091e-02 (0.0e+00)	1.395e-01 (0.0e+00)	4.361e-02 (0.0e+00)	9.091e-02 (0.0e+00)	1.745e-01 (0.0e+00)	1.745e-01 (0.0e+00)	1.389e-01 (2.4e-02)	1.745e-01 (0.0e+00)	1.745e-01 (0.0e+00)	1.745e-01 (0.0e+00)	1.727e-01 (0.0e+00)	1.727e-01 (0.0e+00)
DNR33MS	**6.075e-01 (0.0e+00)**	6.075e-01 (0.0e+00)	5.920e-01 (9.6e-02)	3.896e-01 (2.8e-03)	6.075e-01 (0.0e+00)	4.131e-01 (4.0e-01)	6.028e-01 (1.2e-02)	6.068e-01 (4.5e-03)	6.075e-01 (9.6e-04)	5.192e-01 (4.4e-02)	6.075e-01 (0.0e+00)	6.075e-01 (0.0e+00)	6.075e-01 (0.0e+00)	6.065e-01 (0.0e+00)	6.068e-01 (0.0e+00)
DNR33MP	**8.834e-01 (6.6e-03)**	8.834e-01 (6.6e-03)	8.718e-01 (8.0e-02)	8.162e-01 (2.4e-02)	8.834e-01 (0.0e+00)	8.768e-01 (6.6e-03)	8.834e-01 (0.0e+00)	8.834e-01 (0.0e+00)	8.834e-01 (0.0e+00)	8.834e-01 (6.6e-03)	8.834e-01 (0.0e+00)	8.834e-01 (0.0e+00)	8.834e-01 (6.6e-03)	8.801e-01 (6.6e-03)	8.834e-01 (0.0e+00)
SatelliteTTC25	5.317e-01 (3.2e-02)	5.317e-01 (3.2e-02)	5.736e-01 (1.2e-02)	5.479e-01 (2.2e-02)	4.514e-01 (1.1e-02)	3.652e-01 (9.5e-02)	4.150e-01 (9.2e-02)	3.822e-01 (3.4e-02)	4.026e-01 (4.5e-02)	**4.497e-01 (1.4e-02)**	5.034e-01 (6.8e-02)	5.332e-01 (2.2e-02)	5.133e-01 (2.0e-02)	**5.392e-01 (2.5e-02)**	4.638e-01 (1.1e-02)	5.307e-01 (2.2e-02)
Jobshop5	**7.374e-01 (4.3e-03)**	7.336e-01 (2.3e-03)	6.999e-01 (2.1e-02)	7.210e-01 (5.1e-03)	7.114e-01 (2.2e-03)	6.273e-01 (6.6e-02)	7.369e-01 (2.3e-03)	7.228e-01 (4.4e-03)	7.132e-01 (6.4e-03)	7.193e-01 (1.5e-02)	7.316e-01 (1.2e-03)	7.253e-01 (2.3e-03)	7.334e-01 (5.1e-03)	7.342e-01 (3.6e-03)	7.309e-01 (2.0e-03)
SatelliteTTC50	5.065e-01 (3.7e-02)	4.923e-01 (3.0e-02)	4.160e-01 (5.5e-02)	4.918e-01 (3.6e-02)	4.624e-01 (2.3e-02)	4.151e-01 (4.4e-02)	5.292e-01 (2.1e-02)	5.048e-01 (1.5e-02)	5.070e-01 (2.9e-02)	**5.529e-01 (2.8e-02)**	5.425e-01 (1.5e-02)	5.257e-01 (2.4e-02)	5.123e-01 (1.9e-02)	4.866e-01 (2.6e-02)	5.187e-01 (1.2e-02)
SiteSelect	3.971e-01 (1.3e-03)	4.012e-01 (4.9e-03)	3.785e-01 (4.8e-03)	3.663e-01 (9.3e-03)	3.787e-01 (8.1e-03)	3.986e-01 (3.1e-03)	4.027e-01 (5.9e-03)	4.088e-01 (4.2e-03)	4.104e-01 (1.2e-03)	4.041e-01 (4.0e-03)	4.127e-01 (4.8e-03)	4.119e-01 (5.0e-03)	**4.142e-01 (2.5e-03)**	3.993e-01 (4.6e-03)	4.013e-01 (6.4e-03)
BlendPool	6.771e-01 (5.6e-04)	6.759e-01 (5.8e-03)	5.564e-01 (1.0e-01)	6.685e-01 (2.2e-03)	6.273e-01 (1.6e-02)	6.767e-01 (5.4e-04)	6.786e-01 (1.0e-03)	6.751e-01 (8.5e-04)	6.747e-01 (1.1e-03)	6.748e-01 (3.4e-03)	6.781e-01 (1.8e-03)	6.784e-01 (1.3e-03)	6.788e-01 (2.0e-04)	6.762e-01 (5.7e-03)	**6.788e-01 (5.6e-03)**
ArrayNull	4.004e-01 (1.9e-02)	3.898e-01 (1.1e-01)	2.456e-01 (1.6e-01)	3.294e-01 (9.0e-02)	3.197e-01 (2.7e-02)	3.652e-01 (9.5e-02)	4.150e-01 (9.2e-02)	3.822e-01 (3.4e-02)	4.026e-01 (4.5e-02)	**4.497e-01 (1.4e-02)**	4.117e-01 (3.3e-02)	4.002e-01 (2.7e-02)	3.766e-01 (1.7e-02)	4.048e-01 (2.7e-02)	3.818e-01 (2.4e-02)
WindFarm	4.699e-01 (5.6e-01)	3.430e-01 (2.2e-01)	n/a	7.073e-02 (2.1e-01)	0.000e+00 (4.2e-02)	3.010e-01 (2.3e-01)	0.000e+00 (1.9e-01)	1.445e-02 (9.7e-02)	2.439e-01 (4.4e-01)	4.895e-01 (5.7e-01)	3.731e-01 (4.1e-01)	5.408e-02 (1.4e-01)	4.429e-01 (2.6e-01)	**7.380e-01 (6.9e-02)**	6.147e-01 (7.0e-02)
HomotopyPath	6.436e-01 (1.6e-02)	6.738e-01 (2.7e-02)	**8.250e-01 (1.8e-02)**	4.910e-01 (3.9e-02)	7.979e-01 (3.0e-02)	6.664e-01 (1.1e-01)	6.428e-01 (1.1e-02)	6.495e-01 (4.2e-02)	6.595e-01 (2.5e-02)	6.585e-01 (1.6e-02)	6.520e-01 (1.6e-02)	6.703e-01 (4.5e-02)	6.536e-01 (1.8e-02)	6.687e-01 (1.7e-02)	6.647e-01 (1.2e-02)
NestingLayout	7.433e-01 (1.4e-02)	7.336e-01 (1.4e-02)	n/a	6.044e-01 (2.0e-02)	n/a	7.066e-01 (5.9e-02)	4.942e-01 (2.0e-01)	6.873e-01 (1.9e-02)	6.876e-01 (2.5e-02)	7.117e-01 (2.5e-02)	7.304e-01 (1.5e-02)	7.094e-01 (1.4e-02)	7.080e-01 (2.4e-02)	**7.460e-01 (1.8e-02)**	3.604e-01 (3.5e-02)
ReservoirOps	**6.551e-01 (1.2e-03)**	6.542e-01 (3.8e-03)	n/a	5.798e-01 (2.1e-02)	n/a	6.478e-01 (2.8e-03)	5.347e-01 (1.1e-01)	6.498e-01 (1.8e-03)	6.494e-01 (1.9e-03)	6.449e-01 (1.1e-02)	6.549e-01 (9.9e-04)	6.525e-01 (3.0e-03)	6.533e-01 (2.3e-03)	6.538e-01 (4.8e-03)	4.320e-01 (1.2e-01)
EcoDispatch15MF	6.413e-01 (5.0e-03)	6.408e-01 (5.7e-03)	**6.449e-01 (1.6e-03)**	6.169e-01 (6.5e-03)	6.030e-01 (8.1e-03)	6.134e-01 (1.4e-02)	6.069e-01 (2.2e-02)	5.909e-01 (4.3e-02)	5.647e-01 (2.4e-02)	5.855e-01 (1.8e-02)	6.116e-01 (5.2e-03)	6.314e-01 (6.8e-03)	6.410e-01 (9.9e-03)	6.356e-01 (6.3e-03)	6.383e-01 (6.4e-03)
PMSched13	6.594e-01 (1.2e-02)	6.618e-01 (6.9e-03)	5.820e-01 (6.8e-02)	6.556e-01 (5.5e-03)	**6.722e-01 (0.0e+00)**	6.333e-01 (2.6e-02)	6.608e-01 (1.1e-02)	6.663e-01 (4.1e-03)	6.670e-01 (2.8e-03)	6.649e-01 (3.4e-03)	6.684e-01 (1.4e-03)	6.694e-01 (2.1e-03)	6.581e-01 (5.5e-03)	6.608e-01 (9.6e-03)	6.691e-01 (3.4e-03)
Knapsack24	5.810e-01 (3.4e-04)	5.830e-01 (7.5e-05)	5.750e-01 (3.2e-04)	5.708e-01 (5.5e-04)	5.504e-01 (1.3e-03)	4.392e-01 (6.2e-02)	5.822e-01 (1.6e-04)	5.585e-01 (1.2e-02)	5.470e-01 (9.5e-03)	5.718e-01 (4.9e-03)	5.787e-01 (4.7e-04)	5.787e-01 (6.1e-04)	**5.830e-01 (4.8e-04)**	5.825e-01 (8.1e-05)	5.824e-01 (1.5e-04)
BiTree12	6.740e-01 (8.7e-03)	6.829e-01 (1.2e-02)	n/a	6.086e-01 (5.4e-02)	n/a	4.406e-01 (2.2e-01)	n/a	**6.845e-01 (9.3e-03)**	6.716e-01 (1.9e-02)	6.468e-01 (4.7e-02)	6.739e-01 (1.4e-02)	6.631e-01 (1.0e-02)	6.567e-01 (2.1e-02)	6.694e-01 (1.4e-02)	6.218e-01 (1.1e-02)
Net12Sparse	4.600e-01 (1.6e-03)	4.600e-01 (6.2e-04)	n/a	3.286e-01 (3.3e-02)	n/a	3.947e-01 (7.1e-02)	2.707e-01 (1.6e-01)	4.607e-01 (2.4e-04)	4.603e-01 (4.1e-03)	4.542e-01 (4.4e-03)	4.607e-01 (6.6e-04)	4.603e-01 (1.4e-03)	4.600e-01 (3.2e-05)	4.602e-01 (1.5e-03)	**4.610e-01 (6.8e-04)**
WTA66	8.173e-01 (4.0e-03)	8.190e-01 (3.0e-03)	7.978e-01 (1.9e-02)	8.074e-01 (5.8e-03)	7.751e-01 (1.2e-02)	5.777e-01 (1.1e-01)	8.217e-01 (7.4e-04)	8.155e-01 (3.2e-03)	8.170e-01 (5.3e-03)	7.986e-01 (6.



Supplementary Fig. S1: Friedman mean ranks over the 20 problems with the Nemenyi critical difference ($\alpha = 0.05$) for all four metrics: IGD (a), HV (b), IGDX (c) and feasible rate (d). The objective-space metrics (a,b) are much flatter than the two claim-carrying axes of main-paper Fig. 5.



Supplementary Fig. S2: Fair wins per algorithm across the four metrics (feasibility-gated: problem–algorithm pairs whose ten runs are all infeasible do not compete on IGD/IGDX/HV). $n = 10$ seeds per pair; one win per problem and metric.



Supplementary Fig. S3: Budget slices as heat maps, IGDX at 50% (a) and 100% (b) of budget on a shared colour scale; grey cells are run medians of exactly zero (RWMMOP11 covers its discrete archive). The paired scatter view of the same comparison is main-paper Fig. 9c; the first-hit heat maps are main-paper Fig. 9a,b.

Table S6: Friedman mean ranks over the 20 problems, all four metrics, penalised lens (auto-generated from `cd_ranks.csv`; rows sorted by IGDX). Lower is better.

Algorithm	IGD	IGDX	HV	Feas. rate
DCMMEA	3.85	5.33	4.80	9.07
DRCEA	5.88	5.70	6.38	9.07
CMMOCD	7.25	5.80	8.78	9.07
CMMOCEA	7.58	6.47	8.62	9.07
CMDCoEA	7.42	7.40	8.12	7.62
CTAEA	7.95	7.50	7.08	8.43
NSGAIII	5.85	7.55	6.45	9.07
CCMO	5.35	7.58	5.78	9.07
NSGAII	6.05	8.18	5.88	9.07
CoMMEA	10.43	8.28	8.62	3.92
RVEA	11.00	8.85	12.68	5.03
CMMODE	8.00	8.85	8.45	9.07
CMME	8.65	9.18	7.40	9.07
DNNSGAII	11.45	10.80	12.05	9.07
MOEAD	13.30	12.55	8.93	4.25
Nemenyi CD ($\alpha = 0.05$, $N = 20$)	4.80	4.80	4.80	4.80

Table S7: MMI calibration on standard problems (auto-generated from `mmi_study/validation_standard.csv`).

Case	Rows	MMI	Expected structure (literature)
MMF1	10000	0.95	2 PS (branches meet at one point)
MMF2	10000	1.00	2 PS
MMF3	10000	1.00	2 PS (ring-like)
MMF4	10000	2.90	2 PS curves (4 sheets at fold)
MMF5	10000	2.90	2+ PS
MMF6	10000	2.90	2 PS
MMF7	10000	0.95	1 PS curve, self-intersecting PF
MMF8	10000	2.90	2 PS
MMF1_grid	1500	0.95	same as MMF1 analytic
SYM_PART_simple	3600	12.05	9 PS segments
SYM_PART_rotated	3600	10.34	9 PS segments
IDMPM2T1	240	0.99	2-objective distance-min; global equivalent PS limited by design
IDMPM2T2	240	0.99	2-objective distance-min; global equivalent PS limited by design
IDMPM2T3	144	0.32	2-objective distance-min; global equivalent PS limited by design
IDMPM2T4	240	0.99	2-objective distance-min; global equivalent PS limited by design
IDMPM2T1_e	121	0.00	2-objective distance-min; global equivalent PS limited by design
ZDT1_D2	1500	0.00	unique PS (control, expect 0)
toy_ridge	160801	0.00	connected neutral ridge (expect 0)
toy_two_ridge	802	1.00	2 equivalent ridges (expect 1)
toy_three_ridge	1203	2.00	3 equivalent ridges (expect 2)
toy_half_multimodal	601	0.67	left half 2 ridges (expect ˜0.5)